

# HW3L

A Three-Field Hu–Washizu Tetrahedron in Logarithmic Strain

Research note

Alejandro Mota

September 25, 2026

## Abstract

A tetrahedral element for large-deformation deviatoric plasticity is proposed, with the aim of removing the stabilization that the composite-tetrahedron and variational-multiscale lineages carry. Three fields — the motion, a volumetric strain and a pressure — are coupled through a Hu–Washizu functional. The deviatoric response is evaluated pointwise at quadrature, so the constitutive law is an unmodified small-strain algorithm, and both auxiliary fields are discontinuous, so their elimination is element-local. The stored energy must split exactly into volumetric and isochoric parts, which five of the seven materials in Norma do. With both auxiliary fields in one space the functional reduces exactly to mean dilatation, and what is open is the choice of pair: the discrete space for the displacement together with the discrete space for the pressure. Volumetric locking and spurious soft modes are failures of the two hypotheses of Brezzi’s theorem for that pair, the inf-sup condition and coercivity of the deviatoric form on the discretely isochoric subspace. The first is settled by measurement over a mesh sequence against a Taylor–Hood control: of six element-local pairs only the three-dimensional Crouzeix–Raviart pair — quadratic displacement enriched with face and interior bubbles, over a linear discontinuous pressure — is stable, at a constant of 0.2968; its face bubbles are global, and the pair has 2.6 times the degrees of freedom of the composite tetrahedron it is meant to replace. The second holds in elasticity for every conforming pair, and under a plastic tangent an identity bounds its constant below by  $(1 - \beta)/2$  for every pair, so it cannot separate them. The claim of this note is that the pairs differ instead in whether their soft modes are isochoric, and it is supported on the linearized operator: under a plastic tangent the softest modes of a constant-pressure element carry a dilatation of 0.20 to 0.27 that does not decay under refinement, those of the Crouzeix–Raviart pair one that decays as  $h^2$ , and those of Taylor–Hood are dilatation almost entirely; in a confined plastic zone with a varying flow direction the Crouzeix–Raviart pair has no spurious mode among its twenty softest at the finest mesh, and the constant-pressure pairs retain between five and fifteen. On deformed configurations the inf-sup constant decreases with the shear of the deformation and is unchanged by its volume change. A fourteen-point degree-five quadrature rule and a nodal fifteen-node basis are shown sufficient and are recommended; the explicit stable step is 3.3 times smaller than a linear tetrahedron’s. The nonlinear confirmation — the smoothness of plastic strain fields in a solve — remains to be made, on the Simo–Hughes model both codes ship rather than on the additive logarithmic plasticity specified here.

# 1 Problem statement

Tetrahedra are the only practical choice for meshing complex geometry automatically. Their performance in the regimes that matter here has been the obstacle: near-incompressible response at bulk-to-shear ratios  $\kappa/\mu \lesssim 10^4$ , isochoric plasticity, large rotation, and weak shocks in stiff-bulk materials. The target is one spatial formulation covering all four, with the time integrator the only thing that changes between quasi-static and dynamic regimes.

Two established remedies fall short in ways that are instructive, and the diagnosis below is the reason for the choices that follow.

**Constant pressure relieves locking and introduces soft modes.** The composite tetrahedron [12, 17] and its five-field extension [5] mitigate volumetric locking with an element-constant pressure. Foulk III et al. [5] report that this choice — and it is the choice, not the element — produces a deleterious soft mode “common to both the quadratic and composite tetrahedral element with constant pressure formulations,” remedied there by a convex energy penalty. The mechanism is a coercivity failure on the discretely isochoric subspace: an element-constant multiplier cannot see displacement patterns whose volume change varies within the element and averages to zero, so those patterns carry only shear energy. In elasticity every projected-pressure element has such patterns, and a stable pair keeps their energy bounded away from zero; in a confined plastic zone the shear response itself softens along the flow direction and the pattern can become nearly free. §8.4 records that on the quadratic tetrahedron these patterns are not *exactly* zero-energy, and §9 records why that does not settle the question.

**Residual-based stabilization contaminates the deviator.** Variational-multiscale stabilization of linear tetrahedra [13] introduces a subgrid pressure of the form

$$p' = c_p \tau (\kappa \operatorname{Div} \dot{\varphi} - \dot{p}), \quad \tau = c_\tau \Delta t, \quad (1)$$

which is added to the stress before the constitutive update sees it. Two consequences are structural rather than incidental. The spatial solution inherits a dependence on the *time step* through  $\tau$ , so there is no well-defined quasi-static limit —  $\tau$  is whatever  $\Delta t$  happens to be. And a tuned coefficient  $c_\tau$  modulates a term that reaches the deviator, and therefore the plastic return map, which is the observed origin of oscillations in equivalent plastic strain. No choice of  $c_\tau$  removes this, because the contamination path is the formulation.

**The common cause.** Both are failures of the hypotheses of one theorem, and an earlier version of this paragraph compressed the two hypotheses into one. Brezzi’s theorem for the mixed problem asks two things of the pair — the discrete space  $U_h$  for the displacement together with the discrete space  $Q_h$  for the pressure [3, §3.4, §5.2]: the inf-sup condition, that every pressure mode be resisted by some displacement, whose failure is locking on one side and spurious pressure on the other; and ellipticity of the deviatoric form on the discretely isochoric subspace  $\ker \mathbf{G}_h$ , that no displacement the pressure cannot see be free, whose failure is a soft mode. The first is a property of the pair alone and is settled in §8.5. The second depends on the material tangent as well as on the pair, and §2 states what this note claims about it. Neither difficulty is a stabilization problem: the remedy for the first is a pair inside the admissible window, and the second must then be checked where the material can break it.

## 2 The claim

In elasticity the second hypothesis holds for every conforming pair: the deviatoric form is coercive on all of  $U_h$  by Korn's inequality, so it is coercive on  $\ker \mathbf{G}_h$ , and no elastic soft mode is a failure of the theorem. An earlier version of this note argued against a constant pressure on elastic grounds, by counting exact zero-energy modes; §8.4 shows that those modes do not survive assembly, and the argument is withdrawn.

Under plastic loading the consistent  $J_2$  tangent is

$$\mathbb{C}^{\text{ep}} = 2\mu \left( \mathbb{I}^{\text{dev}} - \beta \mathbf{n} \otimes \mathbf{n} \right), \quad \mathbf{n} := \mathbf{s}^{\text{tr}} / \|\mathbf{s}^{\text{tr}}\|, \quad \beta = \frac{2\mu}{2\mu + \frac{2}{3}\mathcal{H}'}, \quad (2)$$

with  $\beta = 1$  for perfect plasticity: the shear stiffness along the flow direction is removed.

Read as a constant, the second hypothesis cannot separate pairs, and an earlier version of this section expected it to. For every  $\mathbf{v} \in H_0^1(B)$ , integration by parts gives  $\int |\text{dev } \boldsymbol{\epsilon}|^2 dV = \frac{1}{2} \int |\text{Grad } \mathbf{v}|^2 dV + \frac{1}{6} \int (\text{Div } \mathbf{v})^2 dV$ , and  $(\mathbf{n} : \boldsymbol{\epsilon})^2 \leq |\text{dev } \boldsymbol{\epsilon}|^2$  since  $\mathbf{n}$  is a unit deviator, so

$$\frac{1}{2\mu} \int \boldsymbol{\epsilon} : \mathbb{C}^{\text{ep}} : \boldsymbol{\epsilon} dV \geq (1 - \beta) \int |\text{dev } \boldsymbol{\epsilon}|^2 dV \geq \frac{1 - \beta}{2} \int |\text{Grad } \mathbf{v}|^2 dV \quad (3)$$

for every conforming pair and every member of its kernel, isochoric or not. For  $\beta < 1$  the coercivity constant on  $\ker \mathbf{G}_h$  is therefore bounded below uniformly in  $h$  and in the pair. For  $\beta = 1$  the continuum operator loses strong ellipticity: a shear band with normal  $\mathbf{e}_2$  and jump  $\mathbf{e}_1$  has  $\boldsymbol{\epsilon} \propto \text{sym}(\mathbf{e}_1 \otimes \mathbf{e}_2)$ , to which (2) with that  $\mathbf{n}$  assigns zero energy, so the constant tends to zero under refinement for every pair as the discrete space resolves narrower bands (§8.7). Neither limit says anything about the pair.

What distinguishes the pairs is what the soft directions are. A member of  $\ker \mathbf{G}_h$  that is nearly isochoric is a shear band the continuum problem also has, and its low energy is physical. A member with pointwise nonzero dilatation of zero element mean has the volumetric energy  $\kappa \int (\text{Div } \mathbf{v})^2$  in the continuum and none in the element; once the tangent removes its shear energy it is free in the element and stiff in the physics. That is the soft mode reported for constant-pressure elements, and whether  $\ker \mathbf{G}_h$  contains such members is a property of the pair. For a pair satisfying the inf-sup condition,  $\ker \mathbf{G}_h$  approximates the space of isochoric displacements at the rate of  $U_h$  [3, Prop. 5.1.3], so the dilatation carried by any of its members vanishes with  $h$ . For an element-constant pressure on a quadratic displacement,  $\ker \mathbf{G}_h$  contains displacements whose dilatation is nonzero pointwise and averages to zero over each element, and their dilatation does not vanish with  $h$ .

The claim is therefore the following. *Under a plastic tangent, every soft mode of the stable pair with a linear discontinuous pressure carries a dilatation  $\int (\text{Div } \mathbf{v})^2 dV / \int |\text{Grad } \mathbf{v}|^2 dV$  that vanishes under refinement, and a constant pressure has soft modes whose dilatation does not; the soft modes reported for constant-pressure elements are the latter, and no stabilization is needed where they are absent.* Both operators are linear at a fixed plastic state, so the claim is tested without a nonlinear solve, and §8.7 tests it. If the dilatation of the soft modes does not separate the pairs, the formulation has no advantage over the constant-pressure elements it is meant to replace.

### 3 Continuum formulation

#### 3.1 Kinematics and the volumetric–isochoric split

A body  $B \subset \mathbb{R}^3$  undergoes a motion  $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t)$  with deformation gradient  $\mathbf{F} := \text{Grad } \boldsymbol{\varphi}$  and Jacobian  $J := \det \mathbf{F} > 0$ . The multiplicative factorization

$$\mathbf{F} = J^{1/3} \bar{\mathbf{F}}, \quad \bar{\mathbf{F}} := J^{-1/3} \mathbf{F}, \quad \det \bar{\mathbf{F}} = 1, \quad (4)$$

separates the two kinematic contributions this formulation treats differently:  $J$  carries everything the mixed treatment exists for, and  $\bar{\mathbf{F}}$  carries everything the plasticity acts on.

The mixed field is a scalar function of  $J$  alone. Let

$$\theta = \theta(J), \quad \theta'(J) > 0, \quad \theta(1) = 0, \quad \theta'(1) = 1, \quad (5)$$

be a strictly increasing volumetric strain measure, normalized to vanish with unit slope in the undeformed state. The normalization has two consequences. It makes  $\kappa$  in (7) the physical bulk modulus for every admissible  $\theta$ , so that two instances of the formulation are calibrated to the same material rather than to two different ones. And since  $\partial J / \partial \mathbf{F} = J \mathbf{F}^{-\top}$ ,

$$\frac{\partial \theta}{\partial \mathbf{F}} = \theta'(J) J \mathbf{F}^{-\top}, \quad (6)$$

so (5) also forces  $\partial \theta / \partial \mathbf{F} = \mathbf{I}$  at  $\mathbf{F} = \mathbf{I}$  — which is what makes the pressure–displacement pairing reduce to  $\int \bar{p} \text{Div } \boldsymbol{\psi} \, dV$  independently of the choice of  $\theta$  (§4.1).

Two choices are used below and are developed in §3.6: the logarithmic measure  $\theta = \log J$ , and  $\theta = J - 1$ . Both satisfy (5), and by Remark 3 they nevertheless give different elements.

#### 3.2 Stored energy

The formulation is developed for a stored energy that splits exactly along (4), with a volumetric part quadratic in  $\theta$ :

$$W(\mathbf{F}) = W^{\text{vol}}(\theta) + W^{\text{iso}}(\bar{\mathbf{F}}), \quad W^{\text{vol}}(\theta) = \frac{\kappa}{2} \theta^2. \quad (7)$$

Both halves of (7) are requirements on the material rather than conventions, they are requirements of different strength, and §3.5 separates them and reports which materials meet them.

Given (7), the volumetric constitutive relation is linear,

$$p = \partial W^{\text{vol}} / \partial \theta = \kappa \theta, \quad (8)$$

and it is this linearity — not the choice of  $\theta$ , and not the logarithmic measure specifically — that makes the discrete pressure equal  $\kappa \bar{\theta}$  pointwise in §4.2. The elimination itself rests on less, as §3.5 shows.

The isochoric response is whatever  $W^{\text{iso}}$  the material supplies. It is evaluated at the pointwise  $\bar{\mathbf{F}}$  and is never projected, which is the property §5 requires. Plasticity acts entirely inside  $W^{\text{iso}}$ , through an isochoric internal variable evolved by a return map; neither (7) nor (8) changes when it does.

### 3.3 Three-field functional

Only the volumetric response is treated mixedly. Introduce an independent volumetric strain  $\bar{\theta}$  and constrain it to  $\theta(\boldsymbol{\varphi})$  through a multiplier  $\bar{p}$ , which at stationarity is the pressure:

$$\begin{aligned} \Pi[\boldsymbol{\varphi}, \bar{\theta}, \bar{p}] := & \int_B W^{\text{vol}}(\bar{\theta}) \, dV + \int_B W^{\text{iso}}(\bar{\mathbf{F}}(\boldsymbol{\varphi})) \, dV + \int_B \bar{p} (\theta(\boldsymbol{\varphi}) - \bar{\theta}) \, dV \\ & - \int_B R \mathbf{B} \cdot \boldsymbol{\varphi} \, dV - \int_{\partial_T B} \mathbf{T} \cdot \boldsymbol{\varphi} \, dS. \end{aligned} \quad (9)$$

The isochoric energy is evaluated at the *pointwise*  $\bar{\mathbf{F}}(\boldsymbol{\varphi})$  — for the Hencky instance of §3.6, at  $\mathbf{e}(\boldsymbol{\varphi})$  — and carries no auxiliary field. This is deliberate and is discussed in Remark 5.

### 3.4 What the functional reduces to

Equation (9) belongs to an established lineage, and identifying that lineage is a precondition for assessing what in the formulation is new.

Put  $\bar{\theta}$  and  $\bar{p}$  in the same discrete space  $Q_h$ , which §4.1 does for reasons of its own. Stationarity in  $\bar{\theta}$  then gives  $\bar{p} = \kappa \bar{\theta}$  *exactly* rather than after a projection, and stationarity in  $\bar{p}$  gives  $\bar{\theta} = \Pi_h \theta$ , the  $L^2$  projection of  $\theta$  onto  $Q_h$ . The constraint term of (9) then vanishes identically:

$$\int_B \bar{p} (\theta - \bar{\theta}) \, dV = \kappa \int_B \bar{\theta} (\theta - \Pi_h \theta) \, dV = 0, \quad (10)$$

because  $\kappa \bar{\theta} \in Q_h$  while  $(\theta - \Pi_h \theta) \perp Q_h$  by the definition of the projection. What is left is a functional in the motion alone,

$$\Pi[\boldsymbol{\varphi}] = \int_B W^{\text{iso}}(\bar{\mathbf{F}}(\boldsymbol{\varphi})) \, dV + \int_B \frac{\kappa}{2} (\Pi_h \theta)^2 \, dV, \quad (11)$$

whose stationarity conditions are exactly the three-field equations of §3.7.

**Remark 1** (This is mean dilatation, and the pressure space is the contribution). Equation (11) is a displacement formulation carrying a projected volumetric strain — the mean-dilatation construction of Simo et al. [16], and the mechanism of the  $\bar{B}$  and  $\bar{F}$  families; Boffi et al. [3, §8.12] set out the relation between reduced integration, projected pressure, and the mixed method it is equivalent to. Three consequences follow.

The reduction is *exact*, not an approximation, and the reason is (10) and nothing else:  $\bar{p}$  lies in  $Q_h$ , and  $(\theta - \Pi_h \theta)$  is orthogonal to  $Q_h$  by the definition of the projection. No property of the logarithmic measure is used, and none of a quadratic  $W^{\text{vol}}$ . An earlier version of this note asserted that the exactness followed from the Hencky measure; that was wrong, and the error mattered, because it made a property of the *discretization* — that the two auxiliary fields share one space — look like a property of the *material*. What the material must actually supply is weaker, and is set out in §3.5: an exact volumetric–isochoric split, without which the subtraction leaves unprojected volumetric content inside the “deviatoric” term, which then locks. The implementation in Norma refuses materials that fail it.

The novelty here is therefore *not the mechanism*. Projecting the volumetric strain is four decades old. What is open, and what §4.1 and §8 are about, is which space to project onto — a question the classical elements answer by defaulting to a single constant per element, and which §8.5 shows is not a free choice.

And it means this formulation shares its mechanism with the  $\bar{F}$  lineage it is meant to improve on. The claim is not that the mechanism is new; it is that the mechanism was never the problem, and that the constant-pressure choice made alongside it was. Whether replacing that choice is enough is what the verification list of §9 is for, and item 4 — the smoothness of plastic strain fields, the failure this formulation is intended to address — remains untested.

### 3.5 What the formulation requires of the material

Section 3.4 shows the reduction is exact whenever  $\bar{p}$  and  $\bar{\theta}$  share a space. That is a statement about the discretization and says nothing about the material. What the material must supply is (7), and the two halves of it are requirements of quite different strength. Separating them matters, because an earlier version of this note ran them together with a third condition that is not a requirement of this section at all.

**An exact volumetric–isochoric split.** This condition is structural. The element evaluates  $W$  at the pointwise deformation, subtracts the pointwise volumetric energy and adds it back at the projected strain, so the subtraction must leave no volumetric remainder:

$$W(\mathbf{F}) = W^{\text{vol}}(J) + W^{\text{iso}}(\bar{\mathbf{F}}), \quad \bar{\mathbf{F}} := J^{-1/3}\mathbf{F}, \quad \det \bar{\mathbf{F}} = 1. \quad (12)$$

Whether a material satisfies (12) is decidable without knowing  $W^{\text{vol}}$ : if the split exists then  $W(\mathbf{F}) - W(\bar{\mathbf{F}}) = W^{\text{vol}}(J) - W^{\text{vol}}(1)$  depends on  $J$  alone, so holding  $J$  fixed while varying the isochoric part must leave the difference unmoved. Column 2 of Table 1 is that test, run over every material Norma implements. Five of the seven satisfy it, four more than the implementation currently admits.

**A quadratic  $W^{\text{vol}}$ .** This condition is a convenience and not a necessity, and a smaller convenience than an earlier version of this note made it. Stationarity in  $\bar{p}$  gives  $\bar{\theta} = \Pi_h \theta$  regardless of the material, and stationarity in  $\bar{\theta}$  then gives

$$\bar{p} = \Pi_h \left( \partial W^{\text{vol}} / \partial \bar{\theta} \right) \quad (13)$$

in general — the projection of a *known* function of the already computed  $\bar{\theta}$ , one more mass-matrix solve with a nonlinear integrand and nothing implicit. When  $W^{\text{vol}}$  is quadratic its derivative is linear,  $\kappa \bar{\theta}$  already lies in  $Q_h$ , the projection in (13) is the identity, and  $\bar{p} = \kappa \bar{\theta}$  holds *pointwise*; otherwise  $\bar{p} = \Pi_h W^{\text{vol}'}(\Pi_h \theta)$  is a doubly projected pressure. The reduction (11) is exact either way, with  $W^{\text{vol}}(\Pi_h \theta)$  in place of the quadratic, and the elimination of §4.2 is linear either way; what the quadratic case gives is the pointwise relation and one projection fewer.

**The logarithmic measure itself.** This condition is required by nothing in the present section. It is required by §5, for an unrelated reason: only in logarithmic strain is the constitutive law algebraically linear, which is what allows a small-strain return map to be reused verbatim.

Table 1: What the materials implemented in Norma satisfy. Column 2 is the spread of  $W(\mathbf{F}) - W(\bar{\mathbf{F}})$  across twenty random isochoric parts at each of four fixed  $J \in [0.85, 1.2]$ , relative to its own magnitude; machine zero means the split (12) is exact. Columns 3 and 4 are the relative residual of a least-squares fit of the extracted  $W^{\text{vol}}(J) = W(J^{1/3}\mathbf{I}) - W(\mathbf{I})$  over  $J \in [0.8, 1.25]$  to a quadratic in each candidate volumetric variable; in both exact cases the fitted coefficient is  $\kappa/2$  to five digits. Two materials admit a quadratic  $W^{\text{vol}}$ , and they disagree on the variable. Script: `prototype/materials.jl`, which needs Norma.

| material                  | split (12)            | $W^{\text{vol}}$ quadratic in           |                                         |
|---------------------------|-----------------------|-----------------------------------------|-----------------------------------------|
|                           | spread at fixed $J$   | $J - 1$                                 | $\log J$                                |
| Linear elastic            | $2.0 \times 10^0$     | —                                       | —                                       |
| Saint-Venant Kirchhoff    | $3.0 \times 10^1$     | —                                       | —                                       |
| Neo-Hookean               | $4.2 \times 10^{-13}$ | $5.9 \times 10^{-2}$                    | $1.3 \times 10^{-1}$                    |
| Reciprocal neo-Hookean    | $4.8 \times 10^{-13}$ | $1.8 \times 10^{-1}$                    | $6.5 \times 10^{-3}$                    |
| Seth–Hill ( $m = n = 1$ ) | $4.3 \times 10^{-13}$ | $1.8 \times 10^{-1}$                    | $6.5 \times 10^{-3}$                    |
| Hencky                    | $1.4 \times 10^{-13}$ | $1.8 \times 10^{-1}$                    | <b><math>1.7 \times 10^{-15}</math></b> |
| $J_2$ , Simo–Hughes       | $5.9 \times 10^{-13}$ | <b><math>5.3 \times 10^{-15}</math></b> | $1.8 \times 10^{-1}$                    |

**The case developed here.** The formulation is developed in detail for materials satisfying (12) *with* a quadratic  $W^{\text{vol}}$ , because that is where the pointwise relation holds and where both shipped materials sit. Table 1 admits exactly two, and they do not agree on the variable: Hencky is quadratic in  $\log J$  to  $1.7 \times 10^{-15}$ , and the Simo–Hughes  $J_2$  model that Norma and Carina both ship is quadratic in  $J - 1$  to  $5.3 \times 10^{-15}$ . Each is exact in its own variable and wrong by 18% in the other, so material and volumetric variable are not independently selectable, and neither choice generalizes to the other material. §3.6 develops both.

The three remaining split materials — neo-Hookean, its reciprocal, and Seth–Hill — are admissible with one projection more (Remark 2) and are not developed here. What the formulation requires of a material is the split, and five of the seven have it.

**Remark 2** (Extending to a non-quadratic  $W^{\text{vol}}$ ). Nothing in §3.3 or §3.4 breaks, and — an earlier version of this remark had this wrong — nothing becomes a local Newton iteration either. Two things change.

First, (21) acquires a second projection:  $\bar{\theta} = \Pi_h \theta$  from one mass-matrix solve as before, then  $\bar{p} = \Pi_h W^{\text{vol}'}(\bar{\theta})$  from a second solve with the same factorization and a nonlinear integrand. The tangent contribution is  $\int W^{\text{vol}''}(\bar{\theta}) \Pi_h(D\theta[\delta\varphi]) \Pi_h(D\theta[\psi]) \, dV$ , a weighted integral against the same constant mass matrix, and the factorization is reused across Newton iterations exactly as in the quadratic case.

Second, the choice of volumetric variable is no longer read off the material. For a quadratic  $W^{\text{vol}}$  the variable is whichever one makes it quadratic, and Table 1 settles it. For a non-quadratic  $W^{\text{vol}}$  any monotone  $\theta(J)$  satisfying (5) is admissible, and by Remark 3 the choice changes the element. That is the actual consequence, and it is a modeling decision rather than a computational one.

**Remark 3** (The volumetric variable is a modeling choice). Projecting  $\log J$  and projecting  $J - 1$  do not give the same element. Measured on one tetrahedron under a displacement quadratic in  $\mathbf{X}$ , with the element mean as the projection and a Hencky material so that both

Table 2: The two instances of (7). Everything above the rule is fixed by the generic development; everything below it is where the instances differ.  $\mathbf{b}^e := \mathbf{F}^e(\mathbf{F}^e)^\top$  is the elastic left Cauchy–Green tensor and  $\bar{\mathbf{b}}^e := J^{-2/3}\mathbf{b}^e$  its isochoric part.

|                                     | Hencky                                                                                                | Simo–Hughes                                        |
|-------------------------------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| volumetric variable $\theta$        | $\log J$                                                                                              | $J - 1$                                            |
| $\theta'(J)$                        | $1/J$                                                                                                 | 1                                                  |
| $\partial\theta/\partial\mathbf{F}$ | $\mathbf{F}^{-\top}$                                                                                  | $J\mathbf{F}^{-\top}$                              |
| $W^{\text{vol}}(\theta)$            | $\frac{\kappa}{2}(\log J)^2$                                                                          | $\frac{\kappa}{2}(J - 1)^2$                        |
| isochoric argument                  | $\mathbf{e} = \text{dev } \boldsymbol{\epsilon}, \boldsymbol{\epsilon} = \frac{1}{2} \log \mathbf{C}$ | $\bar{\mathbf{b}}^e$                               |
| $W^{\text{iso}}$                    | $\mu \ \mathbf{e}\ ^2$                                                                                | $\frac{\mu}{2}(\text{tr } \bar{\mathbf{b}}^e - 3)$ |
| return map                          | radial return in $\boldsymbol{\epsilon}$                                                              | radial return on $\bar{\mathbf{b}}^e$              |
| small-strain models port verbatim   | yes                                                                                                   | no                                                 |
| shipped in Norma and Carina         | elasticity only                                                                                       | yes, with consistent tangent                       |

variants use the same exact  $W^{\text{vol}}$ , the element energies differ by  $9.4 \times 10^{-4}$  at 0.5% nominal strain,  $1.0 \times 10^{-2}$  at 5%,  $5.6 \times 10^{-2}$  at 20% and  $2.7 \times 10^{-1}$  at 50% (`prototype/materials.jl`). The difference is  $O(\text{strain})$  and vanishes with it, so both are consistent discretizations of the same continuum problem; they are simply different elements, and they differ by tens of percent in exactly the finite-deformation regime this formulation exists to address. Which variable is projected is therefore a modeling decision, not a change of notation.

### 3.6 The two admissible instances

Table 1 leaves two materials. They agree on everything §3.1–§3.4 requires and differ in every instance-specific quantity the formulation touches; Table 2 collects both columns.

The Hencky instance is the one (14) defines: with

$$\boldsymbol{\epsilon} := \frac{1}{2} \log \mathbf{C} \in \mathbb{S}, \quad \mathbf{C} := \mathbf{F}^\top \mathbf{F}, \quad (14)$$

evaluated by inverse scaling-and-squaring with Padé approximants, the volumetric–deviatoric decomposition

$$\boldsymbol{\epsilon} = \frac{1}{3}\theta\mathbf{I} + \mathbf{e}, \quad \theta = \text{tr } \boldsymbol{\epsilon} = \log J, \quad \mathbf{e} = \text{dev } \boldsymbol{\epsilon}, \quad (15)$$

is orthogonal in the Frobenius inner product, and the split of (7) is inherited from the strain measure rather than imposed on it. The Simo–Hughes instance imposes it instead, through the  $J^{-2/3}$  factor in  $\bar{\mathbf{b}}^e$ ;  $\det \bar{\mathbf{b}}^e = 1$  then makes  $W^{\text{iso}}$  independent of  $J$  by construction. Both are exact, and Table 1 confirms both to machine precision. Three consequences follow.

*The generic development covers both.* Nothing in §3.1–§3.4, and nothing in the discretization of §4.1–§4.2, refers to either column of Table 2. The reduction (11), the pointwise relation  $\bar{p} = \kappa\bar{\theta}$ , and the element-local elimination hold for any  $\theta$  satisfying (5) and any  $W^{\text{iso}}$  satisfying (7).

*The pairing at  $\mathbf{F} = \mathbf{I}$  is instance-independent, and this matters for §8.* By (6) and the normalization (5),  $\partial\theta/\partial\mathbf{F} = \mathbf{I}$  at  $\mathbf{F} = \mathbf{I}$  in both columns, so the pairing that §4.1 reduces to  $\int \bar{p} \text{Div } \boldsymbol{\psi} \, dV$  is the same for both. The inf-sup analysis, and with it the negative stability results of §8.5, therefore transfer between instances unchanged: those results are a property of the pair  $(U_h, Q_h)$ , not an artifact of having measured the logarithmic one.

The instances are not interchangeable in plasticity, and this is the only place they are not. The last two rows of Table 2 are the entire difference, and they follow from (15): only in the logarithmic instance is the constitutive law algebraically linear in the strain that the return map sees, which is what lets a small-strain algorithm be reused without reformulation. §5 develops that instance. Remark 7 records that it is the other one that both codes implement.

### 3.7 Variations

With  $\boldsymbol{\varphi} \in U$ ,  $\bar{\theta}, \bar{p} \in Q$  and test functions  $\boldsymbol{\psi} \in U$  vanishing on  $\partial_{\boldsymbol{\varphi}} B$ ,  $\vartheta, \pi \in Q$ :

$$\begin{aligned} D_{\boldsymbol{\varphi}} \Pi(\boldsymbol{\psi}) &= \int_B \bar{p} \frac{\partial \theta}{\partial \boldsymbol{\varphi}}[\boldsymbol{\psi}] \, dV + \int_B \mathbf{s}(\mathbf{e}(\boldsymbol{\varphi})) : \frac{\partial \mathbf{e}}{\partial \boldsymbol{\varphi}}[\boldsymbol{\psi}] \, dV - \int_B R \mathbf{B} \cdot \boldsymbol{\psi} \, dV - \int_{\partial_T B} \mathbf{T} \cdot \boldsymbol{\psi} \, dS = 0, \\ D_{\bar{\theta}} \Pi(\vartheta) &= \int_B (\kappa \bar{\theta} - \bar{p}) \vartheta \, dV = 0, \\ D_{\bar{p}} \Pi(\pi) &= \int_B (\theta(\boldsymbol{\varphi}) - \bar{\theta}) \pi \, dV = 0. \end{aligned} \tag{16}$$

The kinematic derivatives are

$$\frac{\partial \theta}{\partial \boldsymbol{\varphi}}[\boldsymbol{\psi}] = \theta'(J) J \mathbf{F}^{-\top} : \text{Grad } \boldsymbol{\psi}, \quad \frac{\partial \mathbf{e}}{\partial \boldsymbol{\varphi}}[\boldsymbol{\psi}] = \mathbb{L}^{\text{dev}}(\mathbf{C}) : \text{sym}(\mathbf{F}^{\top} \text{Grad } \boldsymbol{\psi}), \tag{17}$$

with  $\mathbb{L}(\mathbf{C}) := \partial \log \mathbf{C} / \partial \mathbf{C}$  the Fréchet derivative of the matrix logarithm [2] and  $\mathbb{L}^{\text{dev}} := \mathbb{L} - \frac{1}{3} \mathbf{I} \otimes \mathbf{C}^{-1}$ . The volumetric derivative follows from Jacobi's formula and (6). The deviatoric expression is written for the Hencky instance of Table 2; the Simo–Hughes instance replaces it with the variation of  $\bar{\mathbf{b}}^e$ , and nothing outside this equation changes.

### 3.8 Stress reconstruction

Collecting (16)<sub>1</sub> into the form  $\int \mathbf{P} : \text{Grad } \boldsymbol{\psi}$  gives, in general,

$$\mathbf{P} = \theta'(J) J \bar{p} \mathbf{F}^{-\top} + \mathbf{F} (\mathbb{L}(\mathbf{C}) : \mathbf{s}). \tag{18}$$

The deviatoric term is  $\mathbb{L}$  and not  $\mathbb{L}^{\text{dev}}$  applied to  $\mathbf{s}$ . Moving  $\mathbb{L}^{\text{dev}}$  off the strain variation in (17) and onto  $\mathbf{s}$  takes its transpose, which carries the  $\frac{1}{3}$  term onto  $\text{tr } \mathbf{s} = 0$ , and  $\mathbb{L}$  is self-adjoint because  $\log \mathbf{C}$  is the gradient of  $\text{tr}(\mathbf{C} \log \mathbf{C} - \mathbf{C})$ . An earlier version of this equation wrote  $\mathbb{L}^{\text{dev}} : \mathbf{s}$ , which read literally adds  $-\frac{1}{3}(\mathbf{C}^{-1} : \mathbf{s}) \mathbf{F}$ ; that term is not zero, since  $\text{tr } \mathbf{s} = 0$  does not make  $\mathbf{C}^{-1} : \mathbf{s}$  vanish, and it is invisible in elasticity only because (18) then collapses to (19), which is correct.

When the deviatoric response is *elastic*,  $\mathbf{s} = 2\mu \mathbf{e}$  is a function of the current  $\mathbf{C}$  alone through an isotropic law, so  $\mathbf{s}$  is coaxial with  $\mathbf{C}$  at every quadrature point; the off-diagonal blocks of  $\mathbb{L}$  in the principal basis then act on nothing and (18) collapses to the push-forward

$$\mathbf{P} = (\varpi \mathbf{I} + \mathbf{s}_x) \mathbf{F}^{-\top} = \boldsymbol{\tau} \mathbf{F}^{-\top}, \quad \boldsymbol{\tau} := \varpi \mathbf{I} + \mathbf{s}_x = J \boldsymbol{\sigma}, \quad \varpi := \theta'(J) J \bar{p}, \tag{19}$$

in which  $\mathbf{s}_x := \mathbf{R} \mathbf{s} \mathbf{R}^{\top}$  is the *spatial* deviator, with  $\mathbf{R}$  the rotation of the polar decomposition  $\mathbf{F} = \mathbf{R} \mathbf{U}$ . The material  $\mathbf{s}$  of (15) and the spatial  $\mathbf{s}_x$  are different tensors, and an earlier version of this equation used one symbol for both; in material form the same statement reads  $\mathbf{P} = \mathbf{F}^{-\top} \mathbf{s}$ , which agrees with (19) because  $\mathbf{s}$  and  $\mathbf{U}$  commute when coaxial.

The Kirchhoff pressure  $\varpi$  carries the whole of the instance dependence. For the Hencky instance  $\theta'(J)J = 1$  and  $\varpi = \bar{p}$ ; for the Simo–Hughes instance  $\theta'(J)J = J$  and  $\varpi = J\bar{p}$ , giving  $\boldsymbol{\tau} = J\bar{p}\mathbf{I} + \mathbf{s}$  with  $\bar{p} = \kappa(J - 1)$ , which is the Kirchhoff stress of Simo and Hughes [15, BOX 9.1] and is what both codes compute. That the general form reproduces each implementation without adjustment is the check that (5) was normalized correctly.

**Remark 4** (Coaxiality is an elastic property). Equation (19) rests on  $\mathbf{s}$  being coaxial with  $\mathbf{C}$ , which holds only when  $\mathbf{s}$  is a function of the current  $\mathbf{C}$  alone. It fails in two settings, and the second is the one this formulation exists for. With an independent discrete stress field,  $\mathbf{s}$  is tied to  $\mathbf{C}$  only weakly and is not coaxial with it at an arbitrary iterate. With plasticity, the return map (22) evaluates  $\mathbf{s} = 2\mu(\mathbf{e} - \mathbf{e}_{n+1}^p)$  against a *stored* plastic strain  $\mathbf{e}_n^p$ , which is coaxial with the current  $\mathbf{C}$  only under proportional loading; whenever the principal axes rotate — the large-rotation regime of §1 —  $\mathbf{s}$  is not coaxial with  $\mathbf{C}$ , the off-diagonal blocks of  $\mathbb{L}$  do not vanish, and (19) is wrong. In both settings (19) holds at best at a converged proportional state, and assembling the residual from it gives an inconsistent method that appears to work. Equation (18) is the form to implement, and Miehe et al. [9] carry the full projection tensor for exactly this reason. An earlier version of this remark named only the first setting.

**Remark 5.** No deviatoric mixed pair appears because none is needed. Volumetric locking is a constraint phenomenon: at  $\kappa/\mu \gg 1$  the near-isochoric constraint over-determines the displacement space, and relieving it requires a multiplier. The deviatoric response is not constrained in the same sense and admits no analogous locking mode. Adding a deviatoric pair adds degrees of freedom, an inf-sup condition that has to be established separately, and — if strain and stress are placed in different spaces — a constitutive law evaluated across two discretizations, which is the aliasing that corrupts plastic strain fields. §7 gives the disciplined way to add strain as an independent field should a later requirement demand it.

## 4 Discretization

### 4.1 The pair

Let  $U_h \subset U$  carry the motion and  $Q_h \subset Q$  carry both  $\bar{\theta}$  and  $\bar{p}$ . Two requirements fix the choice.

*Same space for both scalars.*  $\bar{\theta}$  and  $\bar{p}$  are placed in one space  $Q_h$ . This is what makes the constraint term vanish identically and the functional reduce to (11); it is the only thing that reduction needs. Given it, and a quadratic  $W^{\text{vol}}$  (§3.5), (16)<sub>2</sub> reads  $\bar{p} = \kappa\bar{\theta}$  pointwise rather than in a projected sense, and the composition  $\bar{p} = \kappa\Pi_{Q_h}(\bar{\theta})$  is well defined. Splitting the two scalars across different spaces makes the constitutive relation hold only after a projection, determines one field only through its projection onto the other’s space, and can leave the discrete system singular.

*Inf-sup stability, in three dimensions.* By (17) the pressure–displacement pairing is  $\int \varpi \mathbf{F}^{-\text{T}} : \text{Grad } \boldsymbol{\psi}$  with  $\varpi = \theta'(J)J\bar{p}$ , and not  $\int \bar{p} \mathbf{F}^{-\text{T}} : \text{Grad } \boldsymbol{\psi}$ ; the two coincide at  $\mathbf{F} = \mathbf{I}$  and nowhere else. The normalization (5) sets  $\theta'(1) = 1$ , so  $\varpi = \bar{p}$  there and the pairing reduces to the divergence pairing  $\int \bar{p} \text{Div } \boldsymbol{\psi} \, dV$  — in both instances of Table 2, which is what makes the stability results of §8.5 a property of the pair rather than of the material. That pairing is the one of incompressible linear elasticity, in which the pressure is the multiplier of the constraint

$\text{Div } \boldsymbol{\varphi} = 0$ , and its discrete inf-sup condition,

$$\inf_{q \in Q_h} \sup_{\mathbf{v} \in U_h} \frac{\int_B q \text{Div } \mathbf{v} \, dV}{\|\mathbf{v}\|_{H^1} \|q\|_{L^2}} \geq \beta_h, \quad (20)$$

is the condition on the pair. The admissible pairs are therefore those for which  $\beta_h$  in (20) is bounded below uniformly in  $h$ , and stability must also hold on the target range of  $\kappa/\mu$  [3].

The distinction that matters is dimensional and is easy to get wrong. The conforming Crouzeix–Raviart pair [4] is, in two dimensions,  $P_2$  enriched with a cubic interior bubble over  $P_1^{\text{disc}}$ ; its three-dimensional counterpart below carries a quartic interior bubble and cubic face bubbles, and the quadrature degree the element needs follows from the quartic. Unenriched  $P_2/P_1^{\text{disc}}$  is stable in two dimensions only on special meshes. In three dimensions Boffi et al. [3, §8.7] state that even a *constant* discontinuous pressure is not controlled by  $P_2$ : “to control piecewise constant pressure, we need some degrees of freedom on the faces,” and the quadratic space has none. The stable three-dimensional choices with a discontinuous pressure therefore require bubble enrichment of the displacement space, and the enrichment has two parts with two jobs: face bubbles control the constant pressure, and an interior bubble lifts it to a linear one [3, Example 8.7.2]. Table 3 lists the candidates, and §8.5 measures every one of them.

| Displacement                                       | Pressure            | 3D stable | Pressure elimination |
|----------------------------------------------------|---------------------|-----------|----------------------|
| $P_1$                                              | $P_0$               | no        | local                |
| $P_2$                                              | $P_0$               | <b>no</b> | local                |
| $P_2 \oplus b_{\text{face}}$                       | $P_0$               | yes       | local                |
| $P_2$                                              | $P_1^{\text{disc}}$ | <b>no</b> | local                |
| $P_2 \oplus b_{\text{int}}$                        | $P_1^{\text{disc}}$ | no        | local                |
| $P_2 \oplus b_{\text{int}} \oplus b_{\text{face}}$ | $P_1^{\text{disc}}$ | yes       | local                |
| $P_2$                                              | $P_1$ continuous    | yes       | global               |

Table 3: Displacement–pressure pairs on tetrahedra, with three-dimensional stability as Boffi et al. [3, §8.7, Example 8.7.2] give it. An earlier version of this table listed  $P_2/P_0$  as stable. Local elimination requires a discontinuous pressure; three-dimensional stability with a discontinuous pressure requires face bubbles, and a linear pressure requires an interior bubble as well.

The pair this formulation *requires* is  $P_2$  enriched with face and interior bubbles over  $P_1^{\text{disc}}$ . The two enrichments are different kinds of object, and an earlier version of this paragraph conflated them. The interior bubble vanishes on the whole element boundary: it is never shared, never reached by a boundary condition, and condenses away, adding no global unknown. A face bubble vanishes on three faces and not on its own, so the two elements meeting there share it; it is a global unknown and does not condense. The enrichment therefore does add global degrees of freedom, which §8.3 counts. What survives is the weaker and sufficient statement: the *pressure* elimination of §4.2 stays element-by-element, because that rests on  $Q_h$  being discontinuous and not on the displacement space at all.

The linear pressure is not forced by the stability results.  $P_2 \oplus b_{\text{face}}/P_0$  is also stable (§8.5), at a higher constant and with the same number of global unknowns, since the interior bubble condenses. The census of §8.2 that argued against a constant pressure applies to the unenriched displacement space and was in any case withdrawn in §8.4. What separates the two pressure spaces is the claim of §2: a constant pressure leaves in  $\ker \mathbf{G}_h$  the element-scale dilatational

modes that a plastic tangent can free, and a linear one does not. §8.7 measures both pairs: under a uniaxial flow direction the lowest plastic modes of  $P_2 \oplus b_{\text{face}}/P_0$  carry a dilatation of 0.26–0.27 that does not decay under refinement, and those of the Crouzeix–Raviart pair one that decays as  $h^2$ . That measurement, and not the counts, is what the linear pressure is for.

Both that pair and the Taylor–Hood control are measured in §8.5: the enriched pair settles on a discrete inf-sup constant of 0.2968 and Taylor–Hood on 0.2216, and every other element-local pair decays. The trade between them is not stability but where the coupling sits — a global pressure that cannot be eliminated element by element against global face bubbles that cannot be condensed — and §10 puts numbers on it.

## 4.2 Local elimination

Because  $Q_h$  is discontinuous, the mass matrix  $[M]_{mn} = \int_B \mu_m \mu_n \, dV$  is block diagonal by element and constant in the reference configuration. One factorization at setup serves every Newton iteration of every step, and

$$M\{\bar{\theta}\} = \int_B \mu_m \theta(\varphi) \, dV, \quad \bar{p} = \kappa \bar{\theta}, \quad (21)$$

is an element-local solve, with  $\theta = \log J$  for a Hencky material and  $\theta = J - 1$  for the Simo–Hughes one (§3.5). The second relation in (21) holds *pointwise*, rather than only after projection, precisely because  $W^{\text{vol}}$  is quadratic; for a material where it is not, it is replaced by a second projection with the same factorization (Remark 2). Substituting  $\bar{p}(\varphi)$  into (16)<sub>1</sub> leaves a displacement-sized nonlinear problem whose tangent retains the sparsity of the displacement mesh — the auxiliary fields never couple two elements. This is the property that keeps the composite tetrahedron’s tangent as sparse as a displacement element’s, and it is preserved here.

**Remark 6.** The locality is a consequence of the discontinuous pressure, not of the Hu–Washizu structure. A conforming auxiliary field — one requiring inter-element continuity — would make  $\bar{p}(\varphi)$  a global function of  $\varphi$ , and every tangent–vector product would then require a global solve with its mass matrix. That is a substantial regression against the composite tetrahedron, whose pressure elimination is element-local, and is the main reason the pressure is kept discontinuous.

## 5 Plasticity

This section develops the *logarithmic* instance of Table 2. It is the only part of the formulation where the choice of instance changes anything, and the reason to prefer this one; what follows does not hold verbatim for the Simo–Hughes column, and Remark 7 records where that leaves the verification program.

The deviatoric response is evaluated at quadrature points of the displacement element, on  $e(\varphi)$  computed there. The plastic strain is carried as a logarithmic strain  $e^p$  and subtracted additively — the construction of Miehe et al. [9], in which a material geometric pre-processing  $C \mapsto \epsilon$  and a post-processing through  $\mathbb{L}(C)$  bracket an unmodified small-strain algorithm. This is a different constitutive model from the multiplicative  $F = F^e F^p$  plasticity of Simo [14], which also preserves the small-strain return map but applies it to the elastic logarithmic strain of a trial elastic state with an exponential update; the two coincide under proportional loading with fixed principal axes and differ when the axes rotate. An earlier version of this section cited

the latter for the former. Because the constitutive law is linear in logarithmic strain, the  $J_2$  return map is the small-strain radial return verbatim:

$$\begin{aligned}
\mathbf{e}^{\text{tr}} &:= \mathbf{e}(\boldsymbol{\varphi}) - \mathbf{e}_n^{\text{p}}, & \mathbf{s}^{\text{tr}} &:= 2\mu \mathbf{e}^{\text{tr}}, \\
\Delta\gamma : \quad \left\| \mathbf{s}^{\text{tr}} \right\| - 2\mu \Delta\gamma - \sqrt{2/3} \left[ \sigma_y + \mathcal{H}' \left( \bar{\epsilon}_n^{\text{p}} + \sqrt{2/3} \Delta\gamma \right) \right] &= 0 \quad \text{if } f^{\text{tr}} > 0, \quad \text{else } \Delta\gamma := 0, \\
\mathbf{e}_{n+1}^{\text{p}} &:= \mathbf{e}_n^{\text{p}} + \Delta\gamma \mathbf{n}, & \mathbf{n} &:= \mathbf{s}^{\text{tr}} / \left\| \mathbf{s}^{\text{tr}} \right\|, & \bar{\epsilon}_{n+1}^{\text{p}} &:= \bar{\epsilon}_n^{\text{p}} + \sqrt{2/3} \Delta\gamma, \\
\mathbf{s} &:= 2\mu (\mathbf{e}(\boldsymbol{\varphi}) - \mathbf{e}_{n+1}^{\text{p}}).
\end{aligned} \tag{22}$$

The consistency condition is a scalar equation in  $\Delta\gamma$ . For linear hardening it closes in one step,  $\Delta\gamma = f^{\text{tr}} / (2\mu + \frac{2}{3}\mathcal{H}'')$ , which is the form both codes implement; for nonlinear hardening it is a scalar Newton iteration [15, BOX 3.2]. An earlier version of this equation wrote the one-step form as if it were general, and omitted the update of  $\bar{\epsilon}^{\text{p}}$ . Internal variables live at quadrature points and are never projected, so no interpolation error enters the history. Frame-indifferent interpolation, when tensorial internal variables must be transferred, follows Mota et al. [10].

Three properties follow. They are the motivation for the formulation rather than incidental consequences of it.

*The constitutive law is untouched.* All finite-deformation content is absorbed into two material-independent geometric transforms: the forward map  $\mathbf{C} \mapsto \boldsymbol{\epsilon}$  and the tangent map  $\mathbb{L}(\mathbf{C})$  [9]. Between them sits an unmodified small-strain algorithm. Any model already written for small strain —  $J_2$ , kinematic hardening, viscoplasticity, damage — ports without reformulation. This is the practical content of preferring a Hu–Washizu structure over one built around a particular stress–strain pairing. Note that this property belongs to the logarithmic measure and not to the Hu–Washizu structure, so it is the one place where §3.5 does *not* apply: an admissible material in the sense of (12) need not have it, and the Simo–Hughes model both codes ship does not (Remark 7).

*Strain and stress never cross discretizations.*  $\mathbf{e}$  is evaluated and  $\mathbf{s}$  returned at the same quadrature point of the same element. There is no projection between the two, so the aliasing that corrupts equivalent plastic strain in stabilized formulations has no mechanism here.

*No stabilization enters the deviator.* There is no subgrid term, no penalty and no tuned coefficient anywhere in the formulation, so nothing of the form (1) can reach the return map.

**Remark 7** (The model implemented in both codes, and its bearing on claim 4). Equation (22) is the return map this formulation specifies. Neither Norma nor Carina implements it. Both ship the Simo–Hughes  $J_2$  model [15]: a neo-Hookean volumetric–isochoric split,  $W = \frac{\kappa}{2}(J-1)^2 + \frac{\mu}{2}(\text{tr } \bar{\mathbf{b}}^e - 3)$ , with the radial return taken on the isochoric elastic left Cauchy–Green tensor rather than on logarithmic strain. It satisfies (12) exactly —  $\det \bar{\mathbf{b}}^e = 1$  makes the isochoric term independent of  $J$  by construction rather than by an identity — and its  $W^{\text{vol}}$  is quadratic in  $J-1$ , so it is admissible here in the variable  $J-1$ .

Two consequences follow. The first is that claim 4 of §9 will be measured on Simo–Hughes unless a logarithmic  $J_2$  is written for the purpose, and the substitution is larger than an elastic split: it exchanges the additive logarithmic model of (22) for multiplicative plasticity with the return map on  $\bar{\mathbf{b}}^e$ . The smoothness of equivalent plastic strain is a question about whether strain and stress cross discretizations, which is independent of the plasticity model carrying them, so this is a reasonable substitution — but it is a substitution of the model, not only of the split.

The second is that the portability argument above is weakened by it. The claim that any small-strain model ports without reformulation is a property of the logarithmic measure, and Simo–Hughes does not have it: its return map is specific to its own structure. The argument stands for the formulation as specified and does not describe what is currently running.

## 6 Dynamics

The spatial formulation is unchanged; the mass matrix enters only the displacement block. Two implementation points deserve care.

*Lumping.* Row-sum lumping of the quadratic tetrahedron produces non-positive vertex masses and cannot be used. Diagonal-scaling (HRZ) lumping [6], which scales the diagonal of the consistent mass to preserve total mass, is positive for any basis, so positivity is not the issue; on quadratic simplices the known disadvantage of HRZ is accuracy, with the vertex masses small. The issue the enrichment adds depends on the basis that carries it, and §8.8 measures both choices. With hierarchical bubbles a rigid translation has zero coefficient on every bubble degree of freedom, so their diagonal mass is inert for translations, yet HRZ’s normalization counts it: the Lagrange masses come out short of the total by the fraction the bubbles absorb, 51% on the reference tetrahedron, and a rigid translation carries half its momentum. The normalization must then run over the Lagrange degrees of freedom alone, with each bubble keeping its own consistent diagonal, which restores the momentum exactly. With a nodal basis on the fifteen nodes of the TETRA15 arrangement — vertices, edge midpoints, face centroids, centroid — the functions form a partition of unity, HRZ conserves momentum without qualification, and the lumped masses are positive on every element measured. Row-sum lumping was also positive in that basis on the three elements measured, unlike for TETRA10, but its positivity is not guaranteed and HRZ is the choice.

*Critical time step.* From the element eigenvalue bound at  $\kappa/\mu = 2$  (§8.8), the Courant number  $c_p \Delta t/h$  is 0.48 for TETRA4, 0.20 for TETRA10 under HRZ, and for the enriched element 0.15 with the nodal basis under HRZ and 0.11 with the hierarchical basis under Lagrange-only HRZ; on a Kuhn tetrahedron of the Freudenthal mesh, 0.28, 0.13, 0.095 and 0.078. The quadratic element therefore reduces the stable step by a factor of 2.4 against the linear one on the same mesh, and the enrichment a further 1.4 with the nodal basis; the earlier estimate of “roughly a factor of two” for the first was measured, not assumed, only now. Against this, a  $P_2$  mesh resolves a given feature with far fewer elements. Whether the trade-off is favorable depends on the problem and should be measured.

## 7 Optional five-field variant

Should an independent deviatoric strain field become necessary — for non-local regularization, or for internal variables requiring their own discrete representation — the systematic extension introduces  $\bar{\epsilon}$  and  $\bar{\mathbf{s}}$  in the same discontinuous space, of the same degree as  $Q_h$ . Sharing the space is not merely convenient: it keeps the return map at shared quadrature points, keeps every elimination element-local, leaves the discrete system non-singular, and — for a linear deviatoric law, and only then — makes the constitutive relation hold pointwise rather than after a projection; with plasticity  $\bar{\mathbf{s}} = \Pi_h \mathbf{s}(\bar{\epsilon})$  is projected, and an earlier version of this section claimed the pointwise property without that qualification. Placing  $\bar{\epsilon}$  and  $\bar{\mathbf{s}}$  in different spaces

— however natural the spaces are individually — forfeits all of these at once. Remark 4 also applies: with an independent  $\bar{s}$ , the residual must be assembled from (18) and not from the push-forward (19).

## 8 Numerical results

Everything below works on linear operators at  $\mathbf{F} = \mathbf{I}$ : the condensed operator (25) for the census and the locking sweep, and the  $H^1$  seminorm for the stability sweep. None needs a solver, a material model, or a mesh reader. The scripts are in `prototype/`, each reports its own result, and `checks.jl` verifies the quadrature, the bubbles, and the conformity of the enriched spaces against values known in advance.

### 8.1 Method

Meshes are generated rather than read, so the refinement sequence is exact. The unit cube is divided into  $N^3$  cubes and each cube into six tetrahedra by the Freudenthal (Kuhn) subdivision — the six permutations  $(a, b, c)$  giving  $\{\mathbf{0}, \mathbf{e}_a, \mathbf{e}_a + \mathbf{e}_b, (1, 1, 1)\}$ . That decomposition is conforming across shared cube faces, which the more obvious five-tetrahedron split is not. Quadratic meshes add edge midpoints in the node order the reference element expects.

For the stability test the displacement is fixed on the entire boundary and three operators are assembled,

$$K_{ij} = \int \text{Grad } N_i : \text{Grad } N_j \, dV, \quad G_{mi} = \int \mu_m \text{Div } N_i \, dV, \quad M_{mn} = \int \mu_m \mu_n \, dV, \quad (23)$$

with  $K$  the  $H^1$  seminorm, the norm on the displacement in (20). The discrete inf-sup constant is  $\beta_h = \sqrt{\lambda_{\min}}$  over the nonzero spectrum of

$$\mathbf{G} \mathbf{K}^{-1} \mathbf{G}^\top \mathbf{q} = \lambda \mathbf{M} \mathbf{q}. \quad (24)$$

Boundary data fixed everywhere always leaves one zero mode, the hydrostatic one, because  $\int \text{Div } \mathbf{v} = 0$  for  $\mathbf{v} \in H_0^1$ ; any zero beyond that is a spurious pressure mode.

### 8.2 Spurious modes on one element

The soft-mode claim reduces, for linear response, to a rank count on a single element, and can be settled before any mesh or solver exists. Condensing the three fields gives

$$\mathbf{K}_e = \mathbf{K}^{\text{dev}} + \kappa \mathbf{G}^\top \mathbf{M}^{-1} \mathbf{G}, \quad M_{mn} = \int_e \mu_m \mu_n \, dV, \quad G_{mi} = \int_e \mu_m (\text{tr } \mathbf{B})_i \, dV. \quad (25)$$

$\mathbf{K}^{\text{dev}}$  annihilates every mode with zero deviatoric strain at quadrature: the six rigid-body modes, plus the volumetric directions the displacement space can produce. The second term has rank equal to the number of those directions the pressure space can see, at most  $m$ . Anything left over carries neither deviatoric nor volumetric energy and is *exactly* zero-energy:

$$n_{\text{spurious}} = \dim \ker \mathbf{K}^{\text{dev}} - 6 - \text{rank } \mathbf{K}^{\text{vol}}. \quad (26)$$

For the quadratic tetrahedron  $\dim \ker \mathbf{K}^{\text{dev}} = 10$ : six rigid modes and four volumetric directions, the latter because  $\text{tr } \boldsymbol{\epsilon}(\boldsymbol{\varphi})$  is a  $P_1$  field. Table 4 reports (26) against the measured spectrum.

| Pressure space      | $m$ | $\dim \ker \mathbf{K}^{\text{dev}}$ | $\text{rank } \mathbf{K}^{\text{vol}}$ | predicted | observed |
|---------------------|-----|-------------------------------------|----------------------------------------|-----------|----------|
| $P_0$ constant      | 1   | 10                                  | 1                                      | <b>3</b>  | <b>3</b> |
| $P_1$ discontinuous | 4   | 10                                  | 4                                      | 0         | 0        |
| $P_2$ discontinuous | 10  | 10                                  | 4                                      | 0         | 0        |

Table 4: Spurious zero-energy modes of the three-field quadratic tetrahedron at  $\kappa/\mu = 10^4$ , on the affine element. Script: `prototype/softmode.jl`.

Three consequences follow.

The constant-pressure element carries three modes that are *exactly* zero-energy, not merely soft. A mode with no energy at all cannot be tolerated, which is why a penalty is unavoidable once that pressure space is chosen — the remedy in Foulk III et al. [5] is forced by the space, not by the element.

A  $P_1$ -discontinuous pressure removes them completely, with no penalty, and leaves exactly the six rigid-body modes. That is an element-level fact and no more: §8.4 and §8.5 show that the same pair locks and is unstable once assembled.

Enriching the pressure further changes nothing: the rank saturates at four, because four is all the displacement space can produce. Over-enrichment is wasteful here rather than harmful, at least in this rank sense.

The count is a property of the affine element. On a curved element, with the midside nodes displaced so that  $\det \mathbf{J}$  ranges over  $[0.55, 1.19]$ , the isoparametric displacement space is not polynomial in  $\mathbf{x}$ , only the dilatation of its four volumetric directions remains free of deviatoric strain,  $\dim \ker \mathbf{K}^{\text{dev}} = 7$ , and the three constant-pressure modes are soft rather than exactly zero: the softest nonzero mode of the constant-pressure element lies one to two orders of magnitude below the softest mode of the linear-pressure element on the same curved element. An earlier version of this section reported the counts as identical at distortions up to 0.12; that came from a four-point rule, which sees only four points of a curved element, and 0.12 folds the element. The conclusion is unchanged, since the assembled behavior of §8.4 already rests on soft rather than exact modes.

**Remark 8.** This settles the *mechanism* and nothing more. Inf-sup stability is a global property of a mesh sequence and is invisible on one element; the rank argument above says which modes exist, not whether the constant in the stability estimate survives refinement. Claim 1 of §9 remains open, and with it the question of whether bubble enrichment is needed in practice.

### 8.3 The pair, settled by counting

Section 8.2 settled which modes exist on one element. Whether the constant in the stability estimate survives refinement is a property of a mesh sequence, and determining it does not require an eigenvalue computation.

Write  $n_u$  for the number of free displacement degrees of freedom and  $n_p$  for the pressure degrees of freedom. The Schur complement of the mixed system is  $\mathbf{S} = \mathbf{G}\mathbf{K}^{-1}\mathbf{G}^\top$ , an  $n_p \times n_p$  matrix of rank at most  $n_u$ . If  $n_p > n_u$  then  $\mathbf{S}$  is singular *by counting*, the pressure space carries modes no displacement can resist, and no inf-sup constant exists — regardless of the values those constants would otherwise take.

Table 5 counts them on a Freudenthal-subdivided cube.

| Pair                                                                 | $N$ | $n_u$   | $n_p$  | $n_p/n_u$    |
|----------------------------------------------------------------------|-----|---------|--------|--------------|
| $P_2/P_0$                                                            | 12  | 36,501  | 10,368 | 0.284        |
| $P_2/P_1^{\text{disc}}$                                              | 12  | 36,501  | 41,472 | <b>1.136</b> |
| $P_2 \oplus b_{\text{int}}/P_1^{\text{disc}}$                        | 12  | 67,605  | 41,472 | 0.613        |
| $P_2 \oplus b_{\text{int}} \oplus b_{\text{face}}/P_1^{\text{disc}}$ | 12  | 127,221 | 41,472 | 0.326        |

Table 5: Degree-of-freedom counts on the unit cube, displacement fixed on the boundary. Script: `prototype/inf-sup.jl`.

The middle row is the result, and the asymptotics behind it are exact rather than suggestive. The quadratic nodes of this mesh family fill the  $(2N + 1)^3$  grid, so with the whole boundary fixed  $n_u = 3(2N - 1)^3 \rightarrow 24N^3$ , while  $n_p = 4 \cdot 6N^3 = 24N^3$  regardless of boundary data. The ratio tends to one. *The pair is critically determined in the limit*, which is precisely the situation a uniform inf-sup condition cannot survive.

**Remark 9** (What the finite- $N$  column does and does not show). The ratios above exceed one at every  $N$  in the table, but that part is an artifact of the boundary condition and should not be read as the result. Fixing the whole boundary removes  $O(N^2)$  displacement unknowns and none of the pressure, so the approach to the limit is from above; fixing only one face gives  $n_u = 6N(2N + 1)^2$  and the approach is from below, with the ratio *under* one at every computable  $N$ . Both tend to one, and it is the limit that carries the argument. The finite- $N$  sign is a property of how the block is held.

**Proposition 1.** *On a tetrahedral mesh family with at least six tetrahedra per vertex asymptotically — the Freudenthal family among them — and the boundary fixed,  $P_2/P_1^{\text{disc}}$  is not inf-sup stable, and no refinement rescues it.*

The restriction to the mesh family is not decoration; an earlier version of this proposition omitted it and was false. On a general tetrahedral mesh with  $V$  vertices and  $T$  tetrahedra, Euler’s relation gives  $E \approx V + T$  edges, so  $n_u \approx 3(2V + T)$ ,  $n_p = 4T$ , and  $n_p/n_u \rightarrow 4/(3(2V/T + 1))$ , which exceeds one only when  $T/V > 6$ . Below that the count refutes nothing. And the general statement is false: Zhang [18] shows  $P_2/P_1^{\text{disc}}$  stable on Powell–Sabin tetrahedral meshes once their known spurious pressure modes are filtered. The general three-dimensional result is Boffi et al. [3, Remark 8.6.2]: with a  $P_k$  displacement, a discontinuous pressure is stable on general meshes only at degree  $k - 3$ , so for  $k = 2$  nothing is admissible without enrichment.

Adding one interior bubble per element supplies  $3n_{\text{el}}$  further displacement degrees of freedom and moves the ratio to 0.613; adding the four face bubbles as well — the enrichment §4.1 actually calls for — supplies  $3n_{\text{f}}$  more, one vector unknown per interior face, and moves it to 0.326. Both clear the bound. This is the quantitative content of the recommendation in §4.1: the enrichment is not a matter of preference; it is what makes the count admissible. It also fixes the number of global unknowns. The interior bubble condenses; the face bubbles are shared and do not. The global displacement count of the full enrichment is therefore  $36,501 + 59,616 = 96,117$ , which is 2.6 times the 36,501 of  $P_2/P_0$  — also the nodal count of the composite tetrahedron on the same TETRA10 mesh — and 2.5 times the 38,698 global unknowns of Taylor–Hood, whose 2,197 vertex pressures cannot be eliminated. The 127,221 in Table 5 is the dimension of the space, not the number of global unknowns.

The count has a lineage that should be named.  $n_p/n_u$  is the constraint count of Nagtegaal et al. [11], whose subject — locking of simplicial elements in the fully plastic range — is this note’s, and its reciprocal  $r = n_u/n_p$  is the constraint ratio of Hughes [7, §4.3.7], with  $r = 3$  the optimum in three dimensions. At  $N = 12$ :  $r = 3.52$  for  $P_2/P_0$ , 0.88 for  $P_2/P_1^{\text{disc}}$ , 1.63 with the interior bubble, and 3.07 for the full enrichment. The stable pair sits at the classical optimum, and the isochoric fraction  $1 - n_p/n_u$  of §8.4 is  $1 - 1/r$ , with  $2/3$  the optimum. Whether that limiting degeneracy is the practically decisive objection is a separate question, and §8.4 answers it differently than expected.

## 8.4 Assembled behavior, and a correction

The element census of §8.2 is a local statement. An element-local mode survives assembly only if its trace is compatible with every neighbor across every shared face, so the global count can be far smaller than  $3n_{\text{el}}$ , and whether it is bears directly on the argument against a constant pressure, which rests on that count.

Table 6 assembles  $\mathbf{K}^{\text{dev}} + \kappa \mathbf{G}^T \mathbf{M}^{-1} \mathbf{G}$  over meshes under two boundary conditions: the whole boundary fixed, and only  $z = 0$  fixed. The second matters because it is far closer to the confined-but-not-clamped conditions under which soft modes are reported.

The locking measure is again a rank. A displacement mode has volumetric energy exactly when it lies outside  $\ker \mathbf{G}$ , so as  $\kappa \rightarrow \infty$  the reachable deformations collapse onto that kernel and  $\dim \ker \mathbf{G} = n_u - \text{rank } \mathbf{G}$  is the discretely isochoric subspace. A large fraction means the element still deforms; a collapse toward zero is volumetric locking.

The fraction is read against a bound rather than against intuition. Since  $\text{rank } \mathbf{G} \leq n_p$ , it cannot fall below  $1 - n_p/n_u$ . A pair that *attains* that bound spends every pressure unknown on a distinct displacement direction and wastes none; one well above it is leaving pressure modes unresisted; one that cannot reach it, because  $n_p > n_u$ , is over-determined. The difference  $n_p - \text{rank } \mathbf{G}$  counts the pressure modes no displacement resists, which for an all-Dirichlet boundary should be exactly one, the hydrostatic mode; Table 7 carries it as a column.

Three findings, and the first and third correct this note.

**The constant-pressure soft modes do not survive assembly.** Zero spurious modes, under both boundary conditions, at every mesh. Nor are they present as merely low-energy ones: the smallest eigenvalue relative to the deviatoric scale is  $2.39 \times 10^{-3}$  for  $P_0$  against  $2.74 \times 10^{-3}$  for  $P_1^{\text{disc}}$  on the same mesh — indistinguishable, and both reflecting the long-wavelength modes of a lightly constrained block. (An earlier version quoted  $1.51 \times 10^{-3}$  and  $1.84 \times 10^{-3}$ ; the scripts weighted the Voigt shear rows by 2 where the engineering-shear energy metric is  $\frac{1}{2}$ , which overstates shear stiffness by four. No count, rank or inf-sup constant in this note depends on that metric; the two eigenvalues do, and were regenerated.) The three modes per element of §8.2 are real but element-local, and their traces do not match across faces.

So the *exact-zero* soft modes of a constant pressure do not survive assembly on these elements, and the earlier framing here overstated that case. What this test cannot see is the mode Foulk III et al. [5] actually report, which is remedied there by a penalty and therefore has energy — too little, not none. Those are the displacement modes in  $\ker \mathbf{G}_{P_0}$  whose divergence is nonzero pointwise, hundreds per mesh, which carry shear energy alone in the element; whether they are spurious is the coercivity question of §2, which §8.7 measures, and not a count of zeros. The composite tetrahedron of Foulk III et al. [5] is built on a subdivided-linear displacement

| BC      | Pair                                     | $N$ | $n_u$ | $n_p$ | rank $\mathbf{G}$ | fraction     | bound |
|---------|------------------------------------------|-----|-------|-------|-------------------|--------------|-------|
| all     | $P_2/P_0$                                | 3   | 375   | 162   | 158               | 0.579        | 0.568 |
|         |                                          | 4   | 1029  | 384   | 380               | 0.631        | 0.627 |
|         | $P_2/P_1^{\text{disc}}$                  | 3   | 375   | 648   | 363               | <b>0.032</b> | —     |
|         |                                          | 4   | 1029  | 1536  | 957               | <b>0.070</b> | —     |
|         | $\oplus b_{\text{int}}$                  | 3   | 861   | 648   | 644               | 0.252        | 0.247 |
|         |                                          | 4   | 2181  | 1536  | 1532              | 0.298        | 0.296 |
|         | $\oplus b_{\text{face}}$                 | 3   | 1185  | 648   | 646               | 0.455        | 0.453 |
|         |                                          | 4   | 3045  | 1536  | 1534              | 0.496        | 0.496 |
|         | $\oplus b_{\text{int}}, b_{\text{face}}$ | 3   | 1671  | 648   | 647               | <b>0.613</b> | 0.612 |
|         |                                          | 4   | 4197  | 1536  | 1535              | <b>0.634</b> | 0.634 |
|         | $P_2 \oplus b_{\text{face}}/P_0$         | 3   | 1185  | 162   | 161               | 0.864        | 0.863 |
|         |                                          | 4   | 3045  | 384   | 383               | 0.874        | 0.874 |
| $z = 0$ | $P_2/P_0$                                | 3   | 882   | 162   | 162               | 0.816        | 0.816 |
|         |                                          | 4   | 1944  | 384   | 384               | 0.802        | 0.802 |
|         | $P_2/P_1^{\text{disc}}$                  | 3   | 882   | 648   | 522               | 0.408        | 0.265 |
|         |                                          | 4   | 1944  | 1536  | 1216              | 0.374        | 0.210 |
|         | $\oplus b_{\text{int}}$                  | 3   | 1368  | 648   | 648               | 0.526        | 0.526 |
|         |                                          | 4   | 3096  | 1536  | 1536              | 0.504        | 0.504 |
|         | $\oplus b_{\text{face}}$                 | 3   | 1962  | 648   | 648               | 0.670        | 0.670 |
|         |                                          | 4   | 4440  | 1536  | 1536              | 0.654        | 0.654 |
|         | $\oplus b_{\text{int}}, b_{\text{face}}$ | 3   | 2448  | 648   | 648               | 0.735        | 0.735 |
|         |                                          | 4   | 5592  | 1536  | 1536              | 0.725        | 0.725 |
|         | $P_2 \oplus b_{\text{face}}/P_0$         | 3   | 1962  | 162   | 162               | 0.917        | 0.917 |
|         |                                          | 4   | 4440  | 384   | 384               | 0.914        | 0.914 |

Table 6: Assembled census at  $\kappa/\mu = 10^4$ . The bound is  $1 - n_p/n_u$  and is undefined where  $n_p > n_u$ . Assembled zero-energy modes were zero in every row. Script: `prototype/locking.jl`.

space rather than  $P_2$ , which is a different kernel and is not modeled here; that lineage’s soft mode may well be specific to it. Nothing above contradicts their observation; it does not reproduce it on a quadratic tetrahedron.

**The unenriched pair locks, and that is the real objection.** With the boundary fixed,  $P_2/P_1^{\text{disc}}$  retains 3–7% of its deformation modes as  $\kappa$  grows, against 58–63% for  $P_0$ . Under lighter support it is 37–41% against 80%. Either way it is the pair that over-constrains, and locking is a far more damaging failure in practice than a spurious mode, because it is silent: the element returns a stiff response.

**The enrichment, assembled rather than extrapolated.** An earlier version of this table did not build the bubble. It added  $3n_{\text{el}}$  to  $n_u$  and carried rank  $\mathbf{G}$  over from the unenriched pair, on the reasoning that the interior bubble vanishes on the element boundary, and reported an isochoric fraction of 0.56–0.62. The reasoning is sound for  $P_0$ , where the bubble columns of  $\mathbf{G}$  are identically zero (Remark 10), and wrong for  $P_1^{\text{disc}}$ , where they are not: the bubble raises

rank  $\mathbf{G}$  from 957 to 1532 at  $N = 4$ , and the isochoric fraction is 0.252–0.298, not 0.56–0.62. The extrapolation was out by a factor of two, and in the optimistic direction.

The full enrichment does reach 0.61–0.63, in line with  $P_0$  and without  $P_0$ ’s element-level degeneracy — so the discarded figure was numerically close to the right answer for a pair it was not describing. The full enrichment attains the bound  $1 - n_p/n_u$  to three digits and the interior-only one to two, meaning no pressure unknown is spent twice; and for the full pair  $n_p - \text{rank } \mathbf{G} = 1$  at every mesh, the hydrostatic mode alone, where  $P_2/P_0$  and the interior-only pair carry four. That, rather than the soft-mode argument, is what the enrichment is for.

## 8.5 Stability, measured against a validated control

The counting argument of §8.3 is *necessary*. Whether it is sufficient is a different question, and the answer is no. Establishing this carefully matters, because an earlier version of this note relied on the count.

Two changes make a long mesh sequence reachable. With  $\mathbf{K} = \mathbf{P}^\top \mathbf{L} \mathbf{L}^\top \mathbf{P}$  from a sparse Cholesky,  $q^\top \mathbf{G} \mathbf{K}^{-1} \mathbf{G}^\top q = \|\mathbf{L}^{-1}(\mathbf{G}^\top q)_{[p]}\|^2$ , so  $\beta_h$  is the smallest *nonzero singular value* of  $\mathbf{W} = \mathbf{L}^{-1} \mathbf{G}^\top \mathbf{M}^{-1/2}$  — never forming  $\mathbf{G} \mathbf{K}^{-1} \mathbf{G}^\top$ , and so never squaring the condition number of the very quantity being measured. And the null dimension is fixed independently as  $n_p - \text{rank } \mathbf{G}$ , so the cut is taken by index rather than by selecting a threshold. In practice the gap is unambiguous:  $O(10^{-15})$  below it against  $O(10^{-2})$  above.

The sweep also carries a *positive* control. A negative control alone —  $P_1/P_0$ , known unstable — cannot detect a test that reports every pair as unstable.

| Pair                                     | $h = 1/2$ | $1/4$  | $1/6$         | $1/8$         | null        | local rate |
|------------------------------------------|-----------|--------|---------------|---------------|-------------|------------|
| $P_2/P_1$ continuous (control)           | 0.1734    | 0.2186 | 0.2214        | <b>0.2216</b> | 1           | −0.003     |
| $P_2/P_0$                                | 0.1001    | 0.0755 | 0.0544        | 0.0418        | 4           | 0.97       |
| $\oplus b_{\text{int}}$                  | 0.1015    | 0.0756 | 0.0544        | —             | 4           | 0.84       |
| $\oplus b_{\text{face}}$                 | 0.4671    | 0.4228 | 0.4060        | <b>0.3967</b> | 1           | 0.076      |
| $P_2/P_1^{\text{disc}}$                  | 0.1132    | 0.0424 | 0.0277        | —             | $n_p - n_u$ | 1.04       |
| $\oplus b_{\text{int}}$                  | 0.0782    | 0.0564 | 0.0406        | —             | 4           | 0.83       |
| $\oplus b_{\text{face}}$                 | 0.0938    | 0.0701 | 0.0497        | —             | 2           | 0.88       |
| $\oplus b_{\text{int}}, b_{\text{face}}$ | 0.2875    | 0.2962 | <b>0.2968</b> | —             | 1           | −0.003     |

Table 7:  $\beta_h$  over a Freudenthal mesh sequence, all-Dirichlet, ordered as the construction of Boffi et al. [3, Example 8.7.2] runs. “null” is  $n_p - \text{rank } \mathbf{G}$ , the pressure modes no displacement resists, which must be one; “local rate” is  $\ln(\beta_N/\beta_{N+1})/\ln((N+1)/N)$  at the last refinement, constant for a genuine power law and shrinking toward zero for a sequence approaching a limit. Script: `prototype/beta.jl`.

The control behaves exactly as it must:  $\beta_h$  rises and settles on 0.2216, flat to three digits over the last three refinements. Taylor–Hood is stable in three dimensions and the test reports it as such, which is what validates reading the remaining rows.

Every row of Table 7 is what Boffi et al. [3] predict, and an earlier version of this section presented them as findings. What the sweep establishes is that the test bench is sound and that the theory applies to this mesh family; what it adds to the theory is the constants. Read

down the table in the order the construction runs.

**$P_2$  does not control a constant pressure in three dimensions.**  $P_2/P_0$  decays at a local rate approaching one, despite clearing the degree-of-freedom count at 0.284, and carries *four* pressure null modes at every mesh where one is allowed: the hydrostatic mode and three exact spurious modes that no quadratic displacement resists. That is Boffi et al. [3, §8.7] verbatim — “to control piecewise constant pressure, we need some degrees of freedom on the faces” — and the three modes are the missing face control made visible.

**The interior bubble cannot supply it.**  $P_2 \oplus b_{\text{int}}/P_0$  reproduces  $P_2/P_0$  to three digits with the same four null modes, for the reason in Remark 10. Over  $P_1^{\text{disc}}$  the interior bubble raises  $\beta_h$  by about half at every mesh and the rate falls from 1.04 to 0.83, but the sequence decays and four null modes remain. This is the row the count could not have predicted: it clears the count at 0.613 (Table 5) and fails the test.

**Face bubbles do.**  $P_2 \oplus b_{\text{face}}/P_0$  drops the null dimension to one at every mesh and its  $\beta_h$  approaches from above with a local rate that falls from 0.16 to 0.08 over the sequence — not the constant rate of a power law, and not yet the plateau of the control either. Five to seven points cannot separate a limit near 0.35 from a slow decay on their own, and the row is reported as *consistent with stable*; its stability rests on Boffi et al. [3, Example 8.7.2], whose Fortin construction is exactly what the face degrees of freedom make possible. Over  $P_1^{\text{disc}}$  the face bubbles alone leave one spurious linear pressure mode per mesh and  $\beta_h$  decays: controlling the linear modes is the interior bubble’s job.

**Both together are the three-dimensional Crouzeix–Raviart pair, and it is stable.**  $\beta_h$  rises monotonically to 0.2968, flat to three digits over the last three refinements, with a local rate of  $-0.003$  — the control’s own signature, and at a *higher* constant than the control’s 0.2216. The pressure null space is one dimensional at every mesh. *The pair §4.1 calls for works; each half of its enrichment does one job, and neither does the other’s.*

**Remark 10** (What separates the two enrichments). The interior bubble cannot act on a constant pressure at all: it vanishes on the whole element boundary, so  $\int_{B_e} \text{Div } \mathbf{b}_{\text{int}} = 0$  and its columns of  $\mathbf{G}$  against a  $P_0$  mode are identically zero. It reaches only the *linear* pressure modes, which is why it changes  $P_2/P_1^{\text{disc}}$  and leaves  $P_2/P_0$  almost alone — the  $P_2 \oplus b_{\text{int}}/P_0$  row of Table 7 sits just above  $P_2/P_0$  with the same null dimension: above, because enlarging  $U_h$  can only raise  $\beta_h$ , and only just, because the bubble’s columns of  $\mathbf{G}$  are zero. The wiring check itself is `checks.jl`, which reads those columns directly. The face bubbles are what supply a displacement response to the pressure jump across a face, which is the mode a discontinuous pressure actually needs resisted.

**Remark 11** (The count is necessary, not sufficient).  $P_2/P_0$  and  $P_2 \oplus b_{\text{int}}/P_1^{\text{disc}}$  are both counterexamples to any reading of §8.3 as having settled the question. Each passes the count with margin — at 0.284 and 0.613 — and each fails the test. A ratio above one *refutes* a pair; a ratio below one establishes nothing, and the two pairs that clear the count by the widest margin are not the one that turns out to work.

Table 8:  $\beta_h$  at  $N = 5$  on deformed configurations, whole boundary fixed, with the local rate over  $N = 4 \rightarrow 5$  in parentheses for the Crouzeix–Raviart pair; the null dimension is one on every row. The  $P_2/P_0$  column gives the pressure null dimension and  $\beta_h$ : its three exact spurious modes exist only on the affine mesh.

| map             | $J$       | $P_2/P_1$ cont. | $P_2 \oplus b_{\text{face}}/P_0$ | $P_2 \oplus b_{\text{int}} \oplus b_{\text{face}}/P_1^{\text{disc}}$ | $P_2/P_0$ (null, $\beta_h$ ) |
|-----------------|-----------|-----------------|----------------------------------|----------------------------------------------------------------------|------------------------------|
| reference       | 1         | 0.221           | 0.413                            | 0.297 (−0.003)                                                       | 4, 0.063                     |
| twist 90°       | 0.99–1.01 | 0.189           | 0.385                            | 0.204 (0.18)                                                         | 1, 0.027                     |
| twist 180°      | 0.98–1.05 | 0.137           | 0.345                            | 0.116 (0.42)                                                         | 1, 0.051                     |
| parabolic shear | 1         | 0.176           | 0.336                            | 0.183 (0.05)                                                         | 1, 0.005                     |
| inflation       | 1.0–2.3   | 0.192           | 0.368                            | 0.276 (0.01)                                                         | 1, 0.004                     |
| compression     | 0.5       | 0.192           | 0.333                            | 0.213 (0.00)                                                         | 4, 0.038                     |

**Remark 12** (Mesh dependence). These results are for the Freudenthal subdivision, a highly structured family, and structured meshes are where pair counterexamples characteristically arise — the  $P_1/P_0$  checkerboard is at its worst on them. A pair can degenerate on a structured mesh and remain well behaved on a general one. The accurate statement is that  $P_2/P_0$ ,  $P_2/P_1^{\text{disc}}$  and  $P_2 \oplus b_{\text{int}}/P_1^{\text{disc}}$  decay *here*; confirming that as a property of the pairs rather than of the mesh needs a sequence of unstructured meshes, which this study does not have. §8.6 adds one data point in the other direction: on the non-affine images of this mesh the  $P_2/P_0$  constant is smaller still. The affirmative result needs no such confirmation, since Boffi et al. [3, Example 8.7.2] establish it on any shape-regular family; what the sweep adds is the constant.

## 8.6 Stability on a deformed configuration

Every measurement above is at  $\mathbf{F} = \mathbf{I}$ . At a finite deformation the pairing is  $\int \varpi \operatorname{div}_{\mathbf{x}} \psi \, dV$  (§4.1), the divergence pairing on the deformed configuration with the pressure scaled by  $1/J$ , and whether  $\beta_h$  stays bounded below under a large non-affine deformation is what the regime of §1 requires. The operators (23) are assembled on the isoparametric image of the quadratic mesh under a prescribed map, with the derivatives, the divergence and the volume taken on the image and the pressure basis on the reference cell as the element uses it, and with the boundary conditions of the reference cube; the scaling by  $1/J$  changes  $\beta_h$  by at most the ratio of extreme Jacobians and is not applied. Four maps of the unit cube, about its center: a twist about the vertical axis by 90° and by 180° over the height, isochoric and non-affine, with shear strains near 2 at the corners of the latter; a parabolic shear  $x_1 = X_1 + \frac{1}{2}X_3^2$ , isochoric; a radial inflation with  $J$  from 1 to 2.3; and an affine compression to  $J = \frac{1}{2}$  as a control, since an affine map changes  $\beta_h$  only through the element aspect ratio. Table 8 gives the result at  $N = 5$  with the local rate over the last refinement. Script: `prototype/deformed.jl`.

Three findings.

**$\beta_h$  is not uniform in the deformation.** It decreases with the local shear of the map for every pair, and most for the Crouzeix–Raviart pair: from 0.297 to 0.204 under the 90° twist, 0.116 under the 180° twist and 0.183 under the parabolic shear, against  $0.221 \rightarrow 0.189, 0.137, 0.176$  for Taylor–Hood and  $0.413 \rightarrow 0.385, 0.345, 0.336$  for the face-bubbled constant-pressure pair. The inflation, which changes  $J$  by a factor of 2.3 but shears little, leaves the constant at

0.276. The affine compression gives 0.213, flat to four digits from  $N = 3$ : the aspect-ratio effect alone, as expected. The dependence is on the shear of the deformation and not on its volume change.

**At a fixed deformation the sequence in  $h$  is consistent with a positive limit.** The local rate of the Crouzeix–Raviart pair falls with  $N$  on every non-affine map: 0.28, 0.22, 0.18 over  $N = 2$  to 5 for the  $90^\circ$  twist, 0.67, 0.53, 0.42 for the  $180^\circ$  twist, and 0.05 for the parabolic shear (through  $N = 5$ ). A power-law decay would hold its rate. The  $180^\circ$  row has not flattened by  $N = 5$ , and its shear strain of about 2 at the corners is beyond what a plastic zone of the benchmark set reaches; the finding is that the constant degrades with the strain, not that it vanishes. At  $N = 6$  the pair gives 0.1975, 0.1088 and 0.182 on the three non-affine maps, local rates of 0.17, 0.35 and 0.04, each below the rate before it.

**The exact spurious modes of  $P_2/P_0$  belong to the affine mesh.** Under every non-affine map the pressure null dimension of  $P_2/P_0$  falls from four to one and  $\beta_h$  falls to between 0.004 and 0.05: the three modes are no longer exactly unresisted but nearly so, and the pair is worse off the reference configuration than on it. This corrects the direction of Remark 12, which allowed that a structured mesh might show the pair at its worst.

## 8.7 Coercivity under a plastic tangent

The claim of §2 is measured as the constant it names. With  $\mathbf{Z}$  a basis of  $\ker \mathbf{G}_h$ ,

$$\lambda_{\min} = \min_q \frac{q^\top \mathbf{Z}^\top \mathbf{K}^{\text{ep}} \mathbf{Z} q}{q^\top \mathbf{Z}^\top \mathbf{K}^{H^1} \mathbf{Z} q}, \quad \mathbf{K}_{ij}^{\text{ep}} = \frac{1}{2\mu} \int \boldsymbol{\epsilon}_i : \mathbb{C}^{\text{ep}} : \boldsymbol{\epsilon}_j \, dV, \quad (27)$$

is the smallest generalized eigenvalue of the deviatoric form under the tangent (2) against the  $H^1$  seminorm, restricted to the discretely isochoric subspace. It is dimensionless, it is taken on the exact kernel rather than through a penalty, and it is the constant of the second hypothesis; (3) says what to expect of it, at least  $(1 - \beta)/2$  for every pair when  $\beta < 1$  and a decay to zero for every pair when  $\beta = 1$ . The quantity that carries the claim is the dilatation of the eigenmodes,

$$r_v(\mathbf{v}) := \frac{\int (\text{Div } \mathbf{v})^2 \, dV}{\int |\text{Grad } \mathbf{v}|^2 \, dV}, \quad (28)$$

zero for an isochoric field. The flow direction is uniform,  $\mathbf{n} = \text{sym}(\mathbf{e}_1 \otimes \mathbf{e}_2)/\sqrt{2}$  or  $\mathbf{n} = \text{dev}(\mathbf{e}_3 \otimes \mathbf{e}_3)/\|\cdot\|$ , which is the most unfavorable choice: the continuum operator then admits shear bands of any width, a decaying eigenvalue is expected of every pair, and only the dilatation of the mode says whether the band is physical. The tangent is taken at  $\beta = 1$  and at  $\beta = 0.9$ , the boundary is fixed everywhere, and each pair is measured elastically ( $\beta = 0$ ) on the same kernel. Script: `prototype/plastic.jl`.

**The bench is checked by an identity.** By the identity behind (3), the elastic form on any kernel satisfies  $\lambda = \frac{1}{2} + \frac{1}{6}r_v$ , so its lowest mode has  $r_v = 0$  and  $\lambda_1 = 0.5$  exactly. Every row of the sweep reproduces this; at  $N = 2$  for  $P_2/P_0$ , where the kernel has no isochoric member,  $\lambda_1 = 0.5024$  with  $r_v = 0.0146$ , as the identity requires. This is also the check on the Voigt metric correction recorded in §8.4, which the earlier metric would have failed.

Table 9: Lowest mode of the deviatoric form on  $\ker \mathbf{G}_h$  under the tangent (2) at  $\beta = 1$ , all-Dirichlet Freudenthal cube: the eigenvalue  $\lambda_1$  of (27) and the dilatation  $r_v$  of its mode, (28), for a shear and a uniaxial flow direction. The elastic lowest mode has  $\lambda_1 = 0.500$  and  $r_v = 0$  on every row, as the identity behind (3) requires. At  $\beta = 0.9$  every  $\lambda_1$  exceeds the floor 0.05 and the dilatations agree with those below to two digits. Script: `prototype/plastic.jl`.

| Pair                                                                 | $N$ | $\mathbf{n} = \text{sym}(\mathbf{e}_1 \otimes \mathbf{e}_2)/\sqrt{2}$ |        | $\mathbf{n} = \text{dev}(\mathbf{e}_3 \otimes \mathbf{e}_3)/\ \cdot\ $ |               |
|----------------------------------------------------------------------|-----|-----------------------------------------------------------------------|--------|------------------------------------------------------------------------|---------------|
|                                                                      |     | $\lambda_1$                                                           | $r_v$  | $\lambda_1$                                                            | $r_v$         |
| $P_2/P_0$                                                            | 3   | 0.109                                                                 | 0.065  | 0.117                                                                  | 0.208         |
|                                                                      | 4   | 0.067                                                                 | 0.034  | 0.106                                                                  | 0.217         |
|                                                                      | 5   | 0.047                                                                 | 0.029  | 0.101                                                                  | <b>0.220</b>  |
| $P_2 \oplus b_{\text{face}}/P_0$                                     | 3   | 0.044                                                                 | 0.044  | 0.082                                                                  | 0.274         |
|                                                                      | 4   | 0.026                                                                 | 0.028  | 0.074                                                                  | 0.268         |
|                                                                      | 5   | 0.017                                                                 | 0.019  | 0.071                                                                  | <b>0.261</b>  |
| $P_2 \oplus b_{\text{int}} \oplus b_{\text{face}}/P_1^{\text{disc}}$ | 3   | 0.074                                                                 | 0.036  | 0.147                                                                  | 0.018         |
|                                                                      | 4   | 0.049                                                                 | 0.019  | 0.139                                                                  | 0.0074        |
|                                                                      | 5   | 0.032                                                                 | 0.0066 | 0.134                                                                  | <b>0.0042</b> |
| $P_2/P_1$ continuous (control)                                       | 3   | 0.056                                                                 | 0.051  | 0.053                                                                  | 0.861         |
|                                                                      | 4   | 0.030                                                                 | 0.029  | 0.026                                                                  | 0.929         |
|                                                                      | 5   | 0.019                                                                 | 0.020  | 0.016                                                                  | <b>0.956</b>  |

**The constant does not separate pairs, as (3) predicts.** Under the shear direction  $\lambda_1$  decays for every pair, the control included, at rates between  $h^{1.5}$  and  $h^2$ : the discrete spaces resolve narrower shear bands, which the continuum operator admits with zero energy. Under the uniaxial direction no shear band has  $\epsilon \propto \mathbf{n}$ , and  $\lambda_1$  converges for the three element-local pairs, to 0.10 for  $P_2/P_0$ , 0.071 for  $P_2 \oplus b_{\text{face}}/P_0$  and 0.13 for the Crouzeix–Raviart pair, while it decays for Taylor–Hood. At  $\beta = 0.9$  every  $\lambda_1$  lies above the floor of 0.05, the smallest being the control’s 0.067.

**The dilatation of the soft modes separates them, in the direction claimed.** Under the uniaxial direction the lowest modes of both constant-pressure pairs carry  $r_v = 0.20$ – $0.27$  at every refinement, with no decay. These are the modes of §2: the element assigns them no energy for a dilatation the continuum resists with  $\kappa$ , and they are the softest modes the element has. The lowest modes of the Crouzeix–Raviart pair carry 0.040, 0.018, 0.0074 and 0.0042 over  $N = 2$  to 5, decaying as  $h^2$ , the rate at which Boffi et al. [3, Prop. 5.1.3] makes  $\ker \mathbf{G}_h$  approximate the isochoric space; its soft modes are physical. Under the shear direction the lowest modes of every pair are shear bands whose dilatation decays, the Crouzeix–Raviart pair’s three to four times smaller than the others’ at  $N = 5$ . Counted over the twenty lowest modes, with a mode called spurious when  $r_v > 0.1$  (the stable pair’s soft modes lie below 0.02 from  $N = 4$  on, the spurious ones at 0.2–0.4): under the uniaxial direction at  $N = 5$  all twenty of each constant-pressure pair’s and of Taylor–Hood’s are spurious, and none of the Crouzeix–Raviart pair’s at any  $N$ ; under the shear direction the twenty lowest of every element-local pair are bands at  $N = 5$ . The spurious modes of a constant pressure are therefore present in both directions but are the *lowest* modes only where the flow admits no band, which

Table 10: Spurious modes among the twenty lowest of the plastic form on  $\ker \mathbf{G}_h$  ( $r_v > 0.1$ ) in a confined plastic zone with a varying flow direction, at  $N = 3, 4, 5$ .

| field       | boundary   | $P_2/P_0$  | $P_2 \oplus b_{\text{face}}/P_0$ | $P_2 \oplus b_{\text{int}} \oplus b_{\text{face}}/P_1^{\text{disc}}$ | $P_2/P_1$ cont. |
|-------------|------------|------------|----------------------------------|----------------------------------------------------------------------|-----------------|
| indentation | all fixed  | 16, 15, 7  | 18, 17, 5                        | 4, <b>0, 0</b>                                                       | 20, 18, 17      |
| indentation | base fixed | 11, 1, 0   | 13, 5, 0                         | <b>0, 0, 0</b>                                                       | 14, 7, 5        |
| twist       | all fixed  | 14, 18, 15 | 18, 13, 8                        | 1, 4, <b>0</b>                                                       | 20, 20, 20      |
| twist       | base fixed | 5, 2, 0    | 13, 7, 4                         | <b>0, 0, 0</b>                                                       | 12, 13, 12      |

is why the next paragraph counts them in a zone with a varying direction rather than reading the lowest three.

**A confined zone with a varying flow direction.** The flow direction and the plastic zone are taken from a prescribed displacement field:  $\mathbf{n} = \text{dev } \boldsymbol{\epsilon}(\mathbf{u}) / \|\text{dev } \boldsymbol{\epsilon}(\mathbf{u})\|$  at each quadrature point and  $\beta = 1$  where  $|\text{dev } \boldsymbol{\epsilon}(\mathbf{u})|$  exceeds 0.35 of its peak,  $\beta = 0$  elsewhere. Two fields: an indentation of the top face,  $u_3 = -\exp(-r^2/w^2) X_3$  with  $r$  the distance from the vertical axis and  $w = \frac{1}{4}$ , whose zone is 51% of the volume with elastic surroundings; and the displacement of the  $90^\circ$  twist of §8.6, whose zone is 97%. Both are run with the whole boundary fixed and with the base alone fixed. Table 10 counts the spurious modes among the twenty lowest. Script: `prototype/plastic_zone.jl`.

With the whole boundary fixed the constant-pressure pairs retain between five and fifteen spurious modes among their twenty softest at  $N = 5$ , and under the twist the softest mode of  $P_2/P_0$  is spurious at every  $N$  up to 4 ( $r_v = 0.22$ – $0.31$ ) and its third at  $N = 5$  (0.20). The Crouzeix–Raviart pair has none from  $N = 4$  on under the indentation and none at  $N = 5$  under the twist, and none at any  $N$  with the base alone fixed. Taylor–Hood keeps seventeen to twenty, and under the twist its softest mode’s dilatation grows with refinement, 0.29, 0.37, 0.66, 0.74. With the base alone fixed the softest modes of every pair are long-wavelength motions of the block and the counts fall for the constant-pressure pairs too, to zero at  $N = 5$  for  $P_2/P_0$  and four for the face-bubbled pair, so the spurious modes are then above the twentieth; they are not absent, as the all-fixed rows show. What claim 4 of §9 adds is whether a nonlinear solve excites them.

**Taylor–Hood is the worst pair on this measure.** Its pressure is continuous, so its kernel is divergence-free only against continuous linear test functions, and under the uniaxial direction its lowest modes are dilatation almost entirely:  $r_v = 0.86, 0.93, 0.96$  at  $N = 3, 4, 5$ , with  $\lambda_1$  decaying to 0.016. Those modes have no counterpart in the continuum, which resists them with  $\kappa$ . §10 named Taylor–Hood as the fallback should the enrichment add too many unknowns; on this evidence it is a fallback for elasticity only.

## 8.8 Quadrature rule and basis

Two implementation choices set the number of operations per element and its explicit stable step, and neither is decided by the stability results: the quadrature rule, and the basis that carries the enriched space. Both are measured on one element. Scripts: `prototype/quadrature.jl` and `prototype/basis.jl`.

Table 11: Quadrature rules on the enriched element: zero-energy modes of the condensed stiffness (six is correct) and of the deviatoric stiffness (ten: six rigid modes and the four conformal fields), and the extreme softening factors against a 216-point rule, on the reference tetrahedron and on a Kuhn tetrahedron of the Freudenthal mesh.  $\kappa/\mu = 10^4$ . Script: `prototype/quadrature.jl`.

| rule             | points | reference              |      |      | Kuhn                   |      |      |
|------------------|--------|------------------------|------|------|------------------------|------|------|
|                  |        | zero( $\mathbf{K}_e$ ) | min  | max  | zero( $\mathbf{K}_e$ ) | min  | max  |
| RFE degree 2     | 4      | 21                     | 0    | 1.30 | 21                     | 0    | 1.30 |
| RFE degree 3     | 5      | 21                     | < 0  | 1.13 | 21                     | < 0  | 1.12 |
| conical degree 3 | 8      | 6                      | 0.10 | 1.14 | 6                      | 0.03 | 1.21 |
| Keast degree 5   | 14     | 6                      | 1.00 | 1.35 | 6                      | 1.00 | 1.35 |
| conical degree 5 | 27     | 6                      | 0.91 | 1.00 | 6                      | 0.88 | 1.00 |
| conical degree 7 | 64     | 6                      | 1.00 | 1.00 | 6                      | 1.00 | 1.00 |

**Quadrature.** The bench integrates the enriched element to degree 6 so that the linear measurements are exact. For the element the criterion is weaker: a rule is admissible when the condensed stiffness  $\mathbf{K}_e = \mathbf{K}^{\text{dev}} + \kappa \mathbf{G}^T \mathbf{M}^{-1} \mathbf{G}$  has exactly six zero-energy modes, and its effect is the softening it introduces, measured as the extreme generalized eigenvalues of  $\mathbf{K}_e(\text{rule})$  against  $\mathbf{K}_e(\text{exact})$  on the complement of the rigid modes, which are 1 for an exact rule. Table 11 gives both for the rules at hand, with a degree-11 rule of 216 points as the reference. The two rules `ReferenceFiniteElements` supplies, four and five points, leave 21 zero modes and are inadmissible. The eight-point conical rule is rank sufficient and softens the element to 0.03–0.10 of its stiffness in the worst direction, which is also inadmissible. Two degree-5 rules are admissible: the 14-point symmetric rule of Keast [8], which integrates every monomial through degree 5 to  $4 \times 10^{-16}$  and over-stiffens the interior bubble’s own block by 35% since its gradient-squared is of degree 6, and the 27-point conical rule, which under-stiffens the same block by 9–12%. Assembled, both give the Crouzeix–Raviart pair’s constant within 1% (0.293 and 0.298 against 0.297 at  $N = 5$ ), a one-dimensional pressure null space, no spurious mode among the twenty lowest under the uniaxial plastic tangent, and the same dilatation of the lowest mode (0.0073, 0.0074 and 0.0074). The 14-point rule is the recommendation: rank sufficient, positive weights, and 4.6 times fewer points than the exact rule, with a stiffer interior bubble.

**Basis.** The space  $P_2 \oplus b_{\text{int}} \oplus b_{\text{face}}$  can be carried by the hierarchical basis of the bench, ten Lagrange functions and five bubbles, or by the nodal basis with unit value at one of fifteen nodes, the `TETRA15` arrangement: vertices, edge midpoints, face centroids, centroid. The space is the same, so every stiffness and stability result is the same; what differs is conditioning, lumping, and what a rigid translation looks like. Table 12 gives the measurements at  $\kappa/\mu = 2$ . The nodal basis conditions the consistent mass eight times better (94 against 778) and the stiffness somewhat better. Under HRZ lumping the hierarchical basis loses 51% of the momentum of a rigid translation unless the normalization runs over the Lagrange functions alone (§4); the nodal basis conserves it without qualification, and its lumped masses are positive on every element measured. The explicit stable step from the element bound, as the Courant number  $c_p \Delta t/h$ , is 0.476 for `TETRA4`, 0.197 for `TETRA10` under HRZ, 0.145 for the enriched element in the nodal basis under HRZ, and 0.114 in the hierarchical basis under Lagrange-only HRZ, with

Table 12: Basis choice on the reference tetrahedron,  $\kappa/\mu = 2$ ,  $\mu = \rho = 1$ : condition numbers of the condensed stiffness on the complement of the rigid modes and of the consistent mass, the smallest lumped mass over  $\rho V$ , the momentum of a unit translation under the lumped mass over  $\rho V$ , and the Courant number  $c_p \Delta t/h$  from the element bound. Script: `prototype/basis.jl`.

| basis, lumping                     | cond $\mathbf{K}_e$ | cond $\mathbf{M}$ | min $m/\rho V$ | momentum | Courant |
|------------------------------------|---------------------|-------------------|----------------|----------|---------|
| hierarchical, consistent           | 797                 | 778               | —              | —        | 0.092   |
| hierarchical, HRZ over all 15      | —                   | —                 | 0.014          | 0.486    | 0.097   |
| hierarchical, HRZ over Lagrange 10 | —                   | —                 | 0.028          | 1.000    | 0.114   |
| nodal, consistent                  | 559                 | 94                | —              | —        | 0.092   |
| nodal, HRZ                         | —                   | —                 | 0.021          | 1.000    | 0.145   |
| TETRA10, HRZ                       | —                   | —                 | 0.028          | 1.000    | 0.197   |
| TETRA10, row sum                   | —                   | —                 | −0.050         | 1.000    | —       |
| TETRA4, row sum                    | —                   | —                 | 0.250          | 1.000    | 0.476   |

the same ordering on the distorted and Kuhn elements. The nodal basis is the recommendation: it is what a node-based degree-of-freedom manager carries without a new kind of unknown, its mass lumps correctly by the standard construction, and it takes a 20–27% larger stable step.

## 8.9 The earlier eigenvalue sweep

Table 13 reports (24) over the same mesh family, with  $P_1/P_0$  included as a control: it is known unstable, so a test that fails to identify it as such is not measuring what it purports to measure.

| Pair                    | $N$ | $n_u$ | $n_p$ | zero modes | $\beta_h$ |
|-------------------------|-----|-------|-------|------------|-----------|
| $P_1/P_0$ (control)     | 2   | 3     | 48    | 45         | 0.333     |
|                         | 3   | 24    | 162   | 138        | 0.229     |
|                         | 4   | 81    | 384   | 303        | 0.165     |
|                         | 5   | 192   | 750   | 558        | 0.129     |
| $P_2/P_0$               | 2   | 81    | 48    | 4          | 0.100     |
|                         | 3   | 375   | 162   | 4          | 0.093     |
|                         | 4   | 1029  | 384   | 4          | 0.076     |
|                         | 5   | 2187  | 750   | 4          | 0.063     |
| $P_2/P_1^{\text{disc}}$ | 2   | 81    | 192   | 111        | 0.113     |
|                         | 3   | 375   | 648   | 285        | 0.057     |
|                         | 4   | 1029  | 1536  | 579        | 0.042     |
|                         | 5   | 2187  | 3000  | 1029       | 0.034     |

Table 13: Numerical inf-sup sweep. Script: `prototype/inf-sup.jl`.

The control behaves as it must:  $\beta_h$  falls steadily and the zero count grows with the mesh, which is the checkerboard instability  $P_1/P_0$  is known for. The zero counts for  $P_2/P_1^{\text{disc}}$  — 111, 285, 579, 1029 — are  $n_p - n_u$  to within a few modes, the rank deficiency §8.3 explains by counting. The dense eigensolver caps the sequence at  $N = 5$ , and an earlier version of this note read the  $P_2/P_0$  decay over that range as possibly pre-asymptotic; §8.5 extends it to  $N = 8$  and

it is not. The one thing in this table that the later sweep did not supersede is the zero count of *four* for  $P_2/P_0$  at every  $N$ : one is the hydrostatic mode, and the other three are exact spurious pressure modes that no  $P_2$  displacement resists, present at every refinement — the missing face control of Boffi et al. [3, §8.7] made visible. Table 7 carries that count for every pair; the face bubbles are what reduce it to one.

## 9 What must be verified

The formulation rests on falsifiable claims. They are stated explicitly so that the first round of numerical work is decisive.

1. **The pair is inf-sup stable in three dimensions, uniformly on  $\kappa/\mu \leq 10^4$ .** *Settled affirmatively for the Crouzeix–Raviart pair, negatively for every other element-local pair (§8.5).* On the mesh family tested  $\beta_h$  decays for  $P_2/P_0$ , for  $P_2/P_1^{\text{disc}}$ , and for  $P_2 \oplus b_{\text{int}}/P_1^{\text{disc}}$ , while  $P_2 \oplus b_{\text{int}} \oplus b_{\text{face}}/P_1^{\text{disc}}$  settles on 0.2968 at a local rate of  $-0.003$ , above the Taylor–Hood control’s 0.2216 and with a one-dimensional pressure null space at every mesh. The face bubbles are the part that matters; the interior bubble alone is not enough (Remark 10). At finite deformation the pairing is  $\int \varpi \mathbf{F}^{-\top} : \text{Grad } \boldsymbol{\psi} \, dV = \int \varpi \text{div}_{\mathbf{x}} \boldsymbol{\psi} \, dV$ , the divergence pairing taken on the deformed configuration (§4.1), and §8.6 measures it on four deformed states:  $\beta_h$  degrades with the shear of the deformation, to 0.204 at a  $90^\circ$  twist and 0.116 at  $180^\circ$ , is unchanged by a volume change of 2.3, and at each fixed deformation the sequence in  $h$  is consistent with a positive limit. Uniformity in the deformation does not hold and should not be claimed; what holds is a constant that decreases with the local shear strain. What remains is an unstructured mesh sequence (Remark 12). Uniformity in  $\kappa/\mu$  needs no separate measurement: for the penalized saddle point, stability of the incompressible limit gives error estimates independent of the penalty [3, §5.5.2, §8.12.1], and every pair satisfying (20) uniformly may be used for nearly incompressible elasticity [3, Remark 8.12.1].
2. **No soft modes.** *Closed in elasticity; in plasticity, supported on the linearized operator and not yet in a nonlinear solve.* Three exact zero-energy modes per element for a constant pressure (§8.2), none of which survive assembly (§8.4) — but an exact-zero count is the wrong test for the mode Foulk III et al. [5] report, which has small energy and not none. Soft modes and locking are the two hypotheses of Brezzi’s theorem rather than two faces of one condition: the inf-sup condition governs locking and spurious pressure, and ellipticity of the deviatoric form on the discretely isochoric subspace  $\ker \mathbf{G}_h$  governs soft modes [3, §3.4.5, Thm. 5.2.5]. For the stable pair the second holds in elasticity by Korn’s inequality, and Boffi et al. [3, Prop. 5.1.3] then makes  $\ker \mathbf{G}_h$  approximate the isochoric space at the rate of  $U_h$ . In plasticity the deviatoric tangent loses its stiffness along the flow direction, and the hypothesis can fail exactly where the mode is reported. An earlier version of this item said that no linear operator could decide the question; that was wrong, since the consistent tangent at a fixed plastic state is a linear operator, and §8.7 measures on it the constant, which (3) shows cannot separate pairs, and the dilatation of the soft modes, which does: under a uniaxial flow direction the constant-pressure pairs’ softest modes carry a dilatation of 0.20–0.27 that does not decay, and the stable pair’s one that decays as  $h^2$ . Claim 4’s benchmark tests the same question in the nonlinear setting, with a flow direction that varies in space.

3. **No volumetric locking.** *Settled for every pair* (§8.4), by measuring the isochoric subspace  $\dim \ker \mathbf{G}$  rather than by running benchmarks:  $P_1^{\text{disc}}$  locks badly,  $P_0$  does not, and the fully enriched pair reaches 0.61–0.63 while attaining the count bound  $1 - n_p/n_u$ . What remains is a benchmark confirmation, since a healthy isochoric fraction is necessary and not obviously sufficient.
4. **Plastic strain fields are smooth.** The direct comparison against stabilized formulations. Equivalent plastic strain in a confined plastic zone should show no oscillation and require no filtering. This will be measured on the Simo–Hughes  $J_2$  model both codes ship rather than on the logarithmic return map of (22), for the reasons in Remark 7.
5. **Element-local elimination.** The assembled tangent must retain displacement-mesh sparsity, confirming the elimination never couples elements.

The benchmark set is the one on which the composite tetrahedron is known to struggle: punch indentation, a nearly-incompressible Cook’s membrane in three dimensions, large-rotation twist of a bar, and a weak shock in a high- $\kappa/\mu$  material. The comparison of record is against the composite tetrahedron on the same meshes.

## 10 Risks

*The pair is settled; the number of its unknowns is now the risk.* The central risk of the previous version — that no element-local pair had been shown to work — is retired:  $P_2 \oplus b_{\text{int}} \oplus b_{\text{face}}$  over  $P_1^{\text{disc}}$  is stable, with a constant above Taylor–Hood’s (§8.5). What that verification also established is the number of unknowns. The face bubbles are shared between neighbors, so they are global unknowns and do not condense; at  $N = 12$  the enriched space carries 96,117 global displacement degrees of freedom against 36,501 for  $P_2/P_0$  — and for the composite tetrahedron — on the same mesh, a factor of 2.6, and 2.5 times Taylor–Hood’s 38,698. Locality survives where the note claimed it — the *pressure* elimination is still element-by-element, since that depends only on  $Q_h$  being discontinuous — but the claim that the enrichment adds no global unknowns was false, and §4.1 has been corrected.

Whether that factor of 2.6 is repaid by the accuracy of a stable pair on coarser meshes is not established, and it is the comparison that now decides whether the formulation is worth preferring to Taylor–Hood, whose pressure is global but whose displacement space is 2.5 times smaller. That is a benchmark question, not an operator question, and it is answered on the benchmark set below rather than by any of the counts above. For plasticity the comparison is already decided against Taylor–Hood by §8.7: its softest modes under a plastic tangent are dilatation almost entirely, so it is a fallback for elasticity only.

*Quadratic tetrahedra require more operations per element.* With the 14-point rule of §8.8 the enriched element evaluates the constitutive law at 14 points on 45 displacement degrees of freedom, against 4 or 5 on 30 for TETRA10 and one on 12 for TETRA4. The claim that fewer of them are needed is plausible and unproven for this problem class. It should be measured on the benchmark set, not assumed.

*The explicit regime is the weakest part.* HRZ lumping in the nodal basis resolves positivity and momentum, and the reduced critical time step is measured: a Courant number of 0.145 against 0.476 for TETRA4 on the same element, a factor of 3.3, of which 2.4 is the quadratic element and 1.4 the enrichment (§8.8). If weak shocks can be run with the implicit dynamic

integrator the objection disappears; if they must be explicit, that factor against the coarser mesh a quadratic element allows needs measuring before commitment.

*The formulation restricts the material.* §3.5 requires an exact volumetric–isochoric split, and of the seven materials Norma implements five qualify; the two that do not, linear elasticity and Saint-Venant–Kirchhoff, are not finite-deformation models in the sense that matters here. An earlier version of this paragraph stated the restriction as narrower — a quadratic volumetric energy, admitting two — on the belief that the other three require a local Newton iteration per element; they require one projection (Remark 2). What remains outside the split altogether is anisotropic or non-separable response, which would need the five-field variant of §7 or a different treatment entirely. The restriction should be stated whenever the formulation is compared against  $\bar{F}$ , which carries no such requirement.

*Nothing here is novel in its parts.* Mixed volumetric treatment [16], logarithmic-strain plasticity [9] and displacement–pressure pairs satisfying (20) uniformly [3] are all established. The contribution, if there is one, is the claim that combining them correctly removes the need for the stabilization that both the composite-tetrahedron and variational-multiscale lineages carry — and that claim carries only the weight of the numerical evidence supporting it. Face-hybridized methods [1] remain the alternative to weigh if it fails.

## References

- [1] M. Abbas, A. Ern, and N. Pignet. A hybrid high-order method for incremental associative plasticity with small deformations. *Computer Methods in Applied Mechanics and Engineering*, 346:891–912, 2019.
- [2] A. H. Al-Mohy, N. J. Higham, and S. D. Relton. Computing the Fréchet derivative of the matrix logarithm and estimating the condition number. *SIAM Journal on Scientific Computing*, 35:C394–C410, 2013.
- [3] D. Boffi, F. Brezzi, and M. Fortin. *Mixed Finite Element Methods and Applications*, volume 44 of *Springer Series in Computational Mathematics*. Springer, 2013.
- [4] M. Crouzeix and P.-A. Raviart. Conforming and nonconforming finite element methods for solving the stationary Stokes equations I. *Revue Française d’Automatique, Informatique et Recherche Opérationnelle*, 7:33–75, 1973.
- [5] J. W. Foulk III, J. T. Ostien, B. Talamini, M. R. Tupek, N. K. Crane, A. Mota, and M. G. Veilleux. Extending a 10-node composite tetrahedral finite element for solid mechanics. *International Journal for Numerical Methods in Engineering*, 122:6939–6967, 2021.
- [6] E. Hinton, T. Rock, and O. C. Zienkiewicz. A note on mass lumping and related processes in the finite element method. *Earthquake Engineering and Structural Dynamics*, 4:245–249, 1976.
- [7] Thomas J. R. Hughes. *The Finite Element Method: Linear Static and Dynamic Finite Element Analysis*. Prentice-Hall, Englewood Cliffs, NJ, 1987.
- [8] Patrick Keast. Moderate-degree tetrahedral quadrature formulas. *Computer Methods in Applied Mechanics and Engineering*, 55(3):339–348, 1986.

- [9] C. Miehe, N. Apel, and M. Lambrecht. Anisotropic additive plasticity in the logarithmic strain space: modular kinematic formulation and implementation based on incremental minimization principles for standard materials. *Computer Methods in Applied Mechanics and Engineering*, 191:5383–5425, 2002.
- [10] A. Mota, W. Sun, J. T. Ostien, J. W. Foulk III, and K. N. Long. Lie-group interpolation and variational recovery for internal variables. *Computational Mechanics*, 52:1281–1299, 2013.
- [11] J. C. Nagtegaal, D. M. Parks, and J. R. Rice. On numerically accurate finite element solutions in the fully plastic range. *Computer Methods in Applied Mechanics and Engineering*, 4:153–177, 1974.
- [12] J. T. Ostien, J. W. Foulk III, A. Mota, and M. G. Veilleux. A 10-node composite tetrahedral finite element for solid mechanics. *International Journal for Numerical Methods in Engineering*, 107:1145–1170, 2016.
- [13] G. Scovazzi, B. Carnes, X. Zeng, and S. Rossi. A simple, stable, and accurate linear tetrahedral finite element for transient, nearly, and fully incompressible solid dynamics: a dynamic variational multiscale approach. *International Journal for Numerical Methods in Engineering*, 106:799–839, 2015.
- [14] J. C. Simo. Algorithms for static and dynamic multiplicative plasticity that preserve the classical return mapping schemes of the infinitesimal theory. *Computer Methods in Applied Mechanics and Engineering*, 99:61–112, 1992.
- [15] J. C. Simo and T. J. R. Hughes. *Computational Inelasticity*, volume 7 of *Interdisciplinary Applied Mathematics*. Springer, New York, 1998.
- [16] J. C. Simo, R. L. Taylor, and K. S. Pister. Variational and projection methods for the volume constraint in finite deformation elasto-plasticity. *Computer Methods in Applied Mechanics and Engineering*, 51:177–208, 1985.
- [17] P. Thoutireddy, J. F. Molinari, E. A. Repetto, and M. Ortiz. Tetrahedral composite finite elements. *International Journal for Numerical Methods in Engineering*, 53:1337–1351, 2002.
- [18] Shangyou Zhang. Quadratic divergence-free finite elements on Powell–Sabin tetrahedral grids. *Calcolo*, 48:211–244, 2011.